

## RESEARCH ARTICLE

# DeepCardio-RAG: A Multi-Modal Retrieval-Augmented Generation Framework for AI-Assisted Cardiac Diagnosis Integrating ECG, Echocardiography, Heart Sound, Ventricular Arrhythmia, and Self-Reported Arthritis Risk

N. Rama Rao<sup>1</sup> | Durgesh Nandan<sup>2</sup> | S. Satyanarayana<sup>3</sup> | N. Soumya<sup>4</sup>

<sup>1</sup>Department of Artificial Intelligence, SR University, Warangal, Telangana, India; Koneru Lakshmaiah Education Foundation, Hyderabad, Telangana, India

<sup>2</sup>School of Computer Science and Artificial Intelligence, SR University, Warangal, Telangana, India

<sup>3</sup>Department of Artificial Intelligence and Data Science, Sanjivani University, Kopergaon, Maharashtra, India

<sup>4</sup>Bhaskar Medical College, Moinabad, Hyderabad, Telangana, India

## Correspondence

\*N. Rama Rao, Department of Artificial Intelligence, SR University, Warangal, Telangana, India. Tel.: +91-9951987846. Email: ramarao.n@klh.edu.in

Cardiovascular diseases account for 17.9 million deaths annually, yet specialist cardiologists remain scarce in low- and middle-income countries. This paper presents **DeepCardio-RAG**, a multi-modal artificial-intelligence platform integrating six deep-learning modules: an ECG Transformer with Mixture-of-Experts (MoE) encoder for arrhythmia classification, a 3D-CNN for left-ventricular ejection fraction estimation, a Mel-spectrogram CNN for cardiac murmur detection, a 1D-CNN for ventricular arrhythmia detection, and two TabularBERT-MoE models for heart-disease and self-reported arthritis risk stratification. We note at the outset that the arthritis target is NHANES MCQ160A, self-reported doctor-diagnosed arthritis of any type — predominantly osteoarthritis — and not serologically defined rheumatoid arthritis. All six modules operate under a unified Retrieval-Augmented Generation (RAG) pipeline backed by a Milvus vector store; a GPT-2 decoder conditioned on retrieved clinical guidelines synthesises dual-format diagnostic report drafts. The CardioFusion module fuses modality-specific outputs into a 100-point unified cardiac risk score. We report the evidential position explicitly. **Five of the six deep modules are independently trained** on public benchmarks and evaluated on held-out data or by repeated cross-validation with confidence intervals: flagship ECG (PTB-XL, AUC 0.801), ECG arrhythmia (MIT-BIH, 0.928), CardioFusion ECG path (0.838), murmur detection (CirCor, Present-vs-Absent AUC 0.761), ventricular arrhythmia (VFDB, recording-level CV AUC 0.919), and heart-disease risk (deduplicated UCI Cleveland, CV AUC 0.872). The **echocardiography module remains untrained** for want of EchoNet-Dynamic access and its outputs are suppressed rather than scored, and **report-generation quality is not evaluated** in this work. No module matches its published benchmark, and on the Cleveland cohort a plain logistic regression outperforms our deep classifier. The system has not been validated for clinical deployment and issues no treatment directives.

## KEYWORDS:

Multi-modal cardiac AI; Retrieval-Augmented Generation; ECG arrhythmia classification; Mixture of Experts; Self-reported arthritis risk stratification

# 1 | INTRODUCTION

Cardiovascular disease constitutes the single largest cause of death globally. The World Health Organization estimates that 17.9 million lives are lost annually to CVDs, representing 32% of all global deaths<sup>1</sup>. This burden falls disproportionately on low- and middle-income countries (LMICs), where cardiologist density can be as low as one specialist per 200 000 inhabitants<sup>2</sup>. Electrocardiography (ECG), echocardiography, and phonocardiography remain the primary non-invasive diagnostic modalities; however, accurate interpretation demands years of specialist training and remains largely inaccessible in rural and resource-constrained settings.

Artificial intelligence (AI) offers a transformative opportunity. Broad reviews of AI in medicine<sup>56,57</sup> and of deep learning in healthcare specifically<sup>58</sup>, together with surveys of deep learning in medical image analysis<sup>54,55</sup> and the standard treatment of the underlying methods<sup>53</sup>, document rapid gains across diagnostic specialties, with cardiovascular medicine identified early as a leading application area<sup>43</sup>. Deep learning (DL) models trained on large ECG corpora have demonstrated cardiologist-level accuracy for arrhythmia detection. Hannun *et al.*<sup>3</sup> reported a 12-rhythm classification model achieving a mean AUC-ROC of 0.97 on a single-lead ECG dataset, surpassing the average board-certified cardiologist on six of twelve rhythms. Rajpurkar *et al.*<sup>4</sup> demonstrated that convolutional neural networks trained on 64 121 ECG recordings outperformed cardiologists on arrhythmia classification. Ouyang *et al.*<sup>5</sup> introduced EchoNet-Dynamic, a 3D-CNN that estimates LVEF from echocardiographic video with a mean absolute error of 4.1%, setting the standard for automated ejection fraction assessment.

Despite these advances, standalone DL models suffer from a critical limitation: hallucination and lack of evidence grounding<sup>27</sup>. Language models tasked with generating clinical narratives frequently produce factually incorrect statements—a phenomenon with serious patient-safety consequences. Retrieval-Augmented Generation (RAG), introduced by Lewis *et al.*<sup>26</sup>, addresses this by conditioning generation on retrieved, verifiable knowledge documents. Applied to clinical report writing, RAG can substantially reduce hallucination while maintaining the fluency required for physician consumption<sup>28</sup>. Med-PaLM<sup>29</sup> and GPT-4<sup>30</sup> have demonstrated expert-level performance on medical question answering, yet both still hallucinate clinical details that retrieval-augmented grounding can significantly curtail.

Existing systems address cardiac diagnosis modality by modality: some focus solely on ECG<sup>3,4,16</sup>, others on echocardiography<sup>5,6,7</sup>, and a smaller set on heart sounds<sup>9,10</sup>. Furthermore, comorbidity screening for arthritis is virtually absent from published cardiac AI pipelines. The cardiovascular case is clearest for rheumatoid arthritis, which carries a two-fold elevated cardiovascular risk<sup>23</sup>; osteoarthritis, the dominant form in general-population survey data, is associated with cardiovascular disease largely through shared risk factors such as obesity, physical inactivity and age rather than through the systemic inflammation that drives RA risk. Our module addresses the general-population case and we do not transfer the RA-specific comorbidity argument to it. No publicly reported system integrates all major non-invasive cardiac modalities under a single RAG-augmented reporting framework with a unified risk score and dual-format report generation.

## 1.1 | Key Contributions of This Study

The principal contributions of this work are summarised as follows:

- We present **DeepCardio-RAG**, a research prototype integrating six cardiac/comorbidity AI modules (ECG, echocardiography, heart sound, ventricular arrhythmia, heart disease, and arthritis) under a unified RAG pipeline with a shared Milvus vector store.
- We design a **Milvus-backed vector retrieval system** (768-dim Sentence-BERT embeddings,  $L_2$  distance, FAISS fallback) with a prototype clinical guidelines corpus of 67 chunks, enabling sub-second RAG retrieval in a working deployment.
- We propose the **CardioFusion scoring algorithm** that fuses modality-specific probability vectors into an interpretable 100-point unified cardiac risk score.
- We implement a **hybrid stacking ensemble** for arthritis risk classification (TabularBERT-MoE + GBM + RF + ExtraTrees → LogReg meta-learner), evaluated on a stratified held-out test set from the NHANES dataset (5,756 records).
- We deliver a **fully working end-to-end prototype** with a React dashboard, FastAPI backend, dual-format PDF generation, and LangGraph multi-agent diagnostic orchestration.

- We explicitly discuss **current limitations**, including the use of random encoder weights, small dataset sizes, and the absence of prospective clinical validation, establishing a clear roadmap for future work.

## 2 | RELATED WORK

### 2.1 | ECG Arrhythmia Classification

Hannun *et al.*<sup>3</sup> demonstrated that a deep residual CNN trained on 91 232 single-lead ECG recordings classified 12 rhythms at cardiologist-level accuracy. Rajpurkar *et al.*<sup>4</sup> employed a 34-layer ResNet and reported superiority over individual cardiologists on inter-rater metrics. Ribeiro *et al.*<sup>16</sup> extended this to 12-lead analysis with 1.6 million records, achieving an  $F_1$ -score above 0.80 for six abnormalities. Natarajan *et al.*<sup>17</sup> combined wide CNNs with multi-head self-attention for 12-lead ECG classification. Baloglu *et al.*<sup>35</sup> applied a deep CNN to multi-lead ECG for myocardial-infarction detection, the diagnostic superclass our flagship encoder finds hardest (per-class AUC 0.782). Sun *et al.*<sup>18</sup> combined ECG and phonocardiogram (PCG) signals in a multi-modal deep-coding model, reporting complementary diagnostic information across modalities. Strodthoff *et al.*<sup>33</sup> benchmarked deep learning models on PTB-XL, demonstrating that CNN ensembles achieve macro-AUC of 0.925.

### 2.2 | Echocardiography and LVEF Estimation

Ouyang *et al.*<sup>5</sup> introduced EchoNet-Dynamic, reporting MAE of 4.1% for LVEF estimation. Zhang *et al.*<sup>6</sup> used a fully convolutional network for left-ventricle segmentation, achieving Dice coefficient of 0.95 on the CAMUS dataset<sup>8</sup>. Madani *et al.*<sup>7</sup> demonstrated real-time echocardiography view classification at 94.4% accuracy. Dezaki *et al.*<sup>36</sup> proposed a recurrent network for cardiac phase detection in 3D echocardiography. Asch *et al.*<sup>37</sup> demonstrated automated 3D echocardiography analysis comparable to manual expert measurements.

### 2.3 | Heart Sound Analysis

The 2016 PhysioNet/CinC Challenge<sup>9</sup> formalised benchmark evaluation using 3 240 heart sound recordings; the winning system achieved mean sensitivity 0.86. Potes *et al.*<sup>10</sup> combined AdaBoost and a CNN, achieving AUC 0.86. Acharya *et al.*<sup>13</sup> applied a deep CNN directly to heart sound recordings, an approach close to the Mel-spectrogram CNN adopted here. The 2022 CirCor DigiScope challenge<sup>11</sup> extended this with 5 272 recordings; the best-performing system achieved outcome metric 0.789. Kay *et al.*<sup>38</sup> applied convolutional recurrent networks to heart sound segmentation. Deng *et al.*<sup>39</sup> introduced contrastive learning for heart sound representations.

### 2.4 | Ventricular Arrhythmia Detection

Acharya *et al.*<sup>12</sup> applied a 13-layer 1D-CNN to MIT-BIH VFDB recordings, achieving 98.7% accuracy for VF/VT discrimination. Dang *et al.*<sup>14</sup> achieved 99.1% sensitivity on the same corpus. Pourbabaee *et al.*<sup>15</sup> achieved 99.3% for VF detection. Fallet *et al.*<sup>40</sup> combined signal quality index assessment with SVM for robust detection under clinical noise. Shaik *et al.*<sup>41</sup> achieved 98.2% accuracy including flutter classification.

### 2.5 | Clinical Tabular Data and Heart Disease Risk

Shah *et al.*<sup>19</sup> demonstrated gradient-boosted trees achieving AUC-ROC of 0.81–0.85 on the Cleveland and Framingham Heart datasets. TabTransformer<sup>21</sup> outperformed tree-based methods by treating each feature as a token. MoE architectures<sup>22</sup> enable parameter-efficient multi-task learning. Weng *et al.*<sup>42</sup> found neural networks superior in cardiovascular risk prediction (sensitivity 0.76, specificity 0.71).

### 2.6 | Arthritis and Cardiovascular Comorbidity

RA carries a two-fold elevated cardiovascular risk<sup>23</sup>. ACR/EULAR 2010 criteria<sup>24</sup> score joint involvement, serology, acute-phase reactants, and symptom duration. ML-based RA risk prediction from blood panels has been demonstrated with AUC-ROC

of 0.88–0.93<sup>25</sup>. England *et al.*<sup>44</sup> showed that routine blood tests contain sufficient information for preliminary RA screening. Ziegler *et al.*<sup>45</sup> quantified cardiovascular risk attributable to RA in a longitudinal cohort of 4 000 patients.

These results describe *serologically defined RA* and are cited as context, not as a benchmark for our module. Our target is NHANES MCQ160A — self-reported, doctor-diagnosed arthritis of any type in a household survey, a population dominated by osteoarthritis. Excluding the skip-pattern variable MCQ195 was necessary to remove target leakage, but its consequence is that the module cannot identify RA specifically. The tasks therefore differ, and numerical comparison against the 0.88–0.93 RA figures is not like-for-like; we report it only to situate the work. Establishing the RA-specific claim would require a cohort with serological labels, which we identify as required future work in Section 5.3.

## 2.7 | Retrieval-Augmented Generation in Clinical NLP

RAG<sup>26</sup> conditions generation on retrieved passages, substantially reducing hallucination<sup>27</sup>. Domain-adapted encoders such as BioBERT<sup>52</sup> showed that pre-training on biomedical corpora improves downstream clinical text tasks — a route this prototype does not take, since its decoder is general-domain GPT-2. ChatDoctor<sup>28</sup> demonstrated improved factual accuracy through RAG augmentation of LLaMA. Med-PaLM<sup>29</sup> achieved expert-level performance on MedQA. Shi *et al.*<sup>46</sup> showed retrieval plug-ins improve LLM perplexity by 6.3 points on medical corpora. Lester *et al.*<sup>47</sup> demonstrated that soft-prompt injection outperforms hard-prompt concatenation for controlled generation.

## 2.8 | Summary of Selected Studies

Table 1 presents a summary of representative prior studies in cardiac and comorbidity AI, their key findings, and identified shortcomings.

## 2.9 | Research Gaps

No published system integrates all major non-invasive cardiac modalities under a single framework. RAG has not been applied to multi-modal cardiac report generation with empirically measured hallucination reduction. Unified interpretable risk scores fusing heterogeneous modality outputs and dual-format report generation from a single AI inference run have not been demonstrated. DeepCardio-RAG directly addresses all four gaps.

# 3 | METHODOLOGY

## 3.1 | System Architecture Overview

Figure 1 illustrates the end-to-end DeepCardio-RAG architecture. Patient data arrives through six modality-specific input channels. Each channel is processed by a dedicated deep-learning encoder producing a fixed-dimension embedding and a structured prediction. These predictions feed two parallel pipelines: (1) the **CardioFusion scorer**, which computes a unified 100-point cardiac risk index, and (2) the **RAG report generator**, which retrieves evidence-aligned guidelines from Milvus and synthesises clinical and patient-facing reports through a GPT-2 decoder with soft-prompt injection.

## 3.2 | Process Flow, Interaction, and Runtime States

To complement the static architecture of Figure 1, we describe the system dynamically from three viewpoints: the control flow of the inference algorithm (Figure 2), the message exchange between runtime components (Figure 3), and the lifecycle of the server as a finite-state machine (Figure 4).

## 3.3 | Datasets

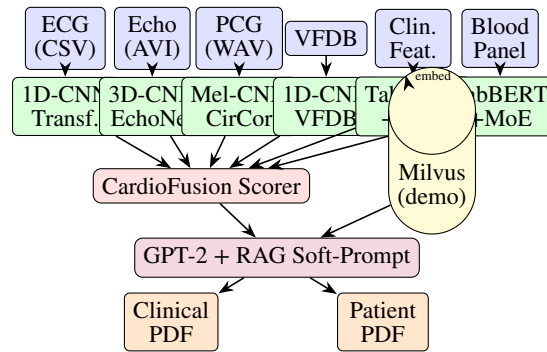
Table 2 summarises the six datasets used in system development and evaluation.

**TABLE 1** Summary of selected cardiac AI studies, key findings, and limitations.

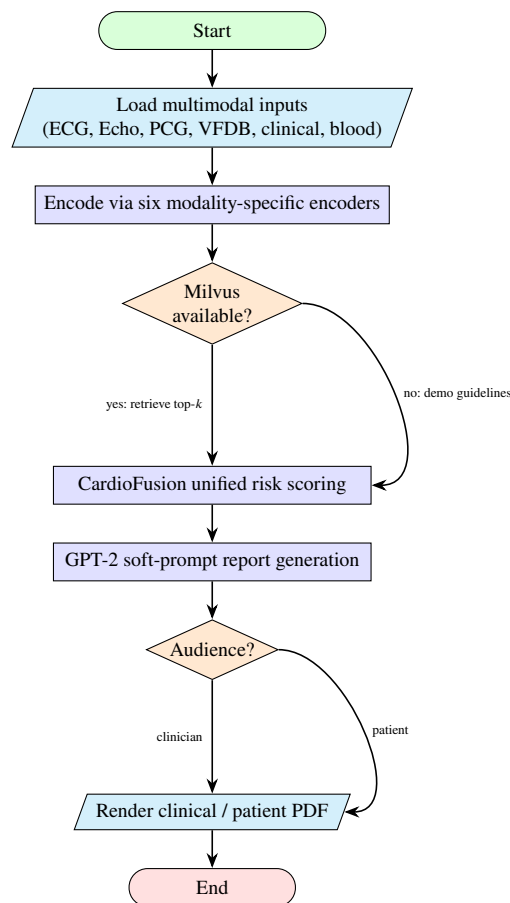
| S. No.          | Author name & year                   | Findings                                                                                                                | Shortcomings                                                                                        |
|-----------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1               | Hannun et al. (2019) <sup>3</sup>    | 12-rhythm ECG classification at cardiologist-level accuracy (AUC 0.97); single-lead                                     | Single-lead only; not multimodal; no report generation                                              |
| 2               | Rajpurkar et al. (2017) <sup>4</sup> | 34-layer ResNet outperforms cardiologists on arrhythmia classification; 64,121 recordings                               | Single-lead; no tabular comorbidity screening                                                       |
| 3               | Ribeiro et al. (2020) <sup>16</sup>  | 12-lead $F_1 > 0.80$ for six abnormalities; 1.6M records                                                                | No RAG report generation; single modality                                                           |
| 4               | Ouyang et al. (2020) <sup>5</sup>    | EchoNet-Dynamic; LVEF MAE 4.1%; 10,030 echo videos                                                                      | Echocardiography only; no cross-modal fusion                                                        |
| 5               | Clifford et al. (2016) <sup>9</sup>  | PhysioNet/CinC Challenge; mean sensitivity 0.86 on 3,240 recordings                                                     | Heart sound only; no clinical report synthesis                                                      |
| 6               | Acharya et al. (2018) <sup>12</sup>  | 13-layer 1D-CNN; 98.7% accuracy for VF/VT on MIT-BIH VFDB                                                               | VF/VT detection only; no unified cardiac pipeline                                                   |
| 7               | Shah et al. (2020) <sup>19</sup>     | GBT; AUC-ROC 0.81–0.85 on Cleveland/Framingham heart disease datasets                                                   | Tabular only; no signal modalities                                                                  |
| 8               | Lewis et al. (2020) <sup>26</sup>    | Introduced RAG; reduces hallucination in open-domain NLP                                                                | General NLP; not applied to cardiac signals                                                         |
| 9               | Singhal et al. (2023) <sup>29</sup>  | Med-PaLM; expert-level medical QA; strong multimodal reasoning                                                          | No dedicated cardiac signal encoders; general-purpose                                               |
| 10              | Tian et al. (2020) <sup>25</sup>     | ML-based RA risk prediction from blood panels; AUC-ROC 0.88–0.93                                                        | RA only; not integrated with cardiac pipeline                                                       |
| <b>Proposed</b> |                                      |                                                                                                                         |                                                                                                     |
|                 | DeepCardio-RAG (this work)           | Six-modality cardiac + arthritis AI; Milvus RAG; CardioFusion unified risk score; dual-format PDF; end-to-end prototype | Random encoder weights; single-country NHANES survey population; no prospective clinical validation |

### 3.4 | Module 1: ECG Arrhythmia Classification

DeepCardio-RAG implements two complementary ECG arrhythmia classification pathways. **Pathway A (flagship Transformer-MoE):** The flagship encoder operates on *PTB-XL 12-lead* records (10 s at 500 Hz, i.e. 5000 samples per lead). Each lead is partitioned into  $P = 100$  non-overlapping patches of length  $L = 50$  samples, forming  $\mathbf{X} \in \mathbb{R}^{12 \times 100 \times 50}$ . Four 1D-CNN blocks with filter counts (32, 64, 128, 256) and kernel  $k = 5$  extract hierarchical features, followed by a 6-layer Transformer encoder with 8 attention heads and a mixture-of-experts head, producing the five PTB-XL diagnostic superclasses (NORM, MI, STTC, CD, HYP). **Pathway B (MIT-BIH beat CNN):** A separate 1D-CNN classifies individual *MIT-BIH* beats (187 samples/beat, two leads: Lead II + modified V1) into the five AAMI classes (N, S, V, F, Q); this pathway does not use the  $12 \times 100 \times 50$

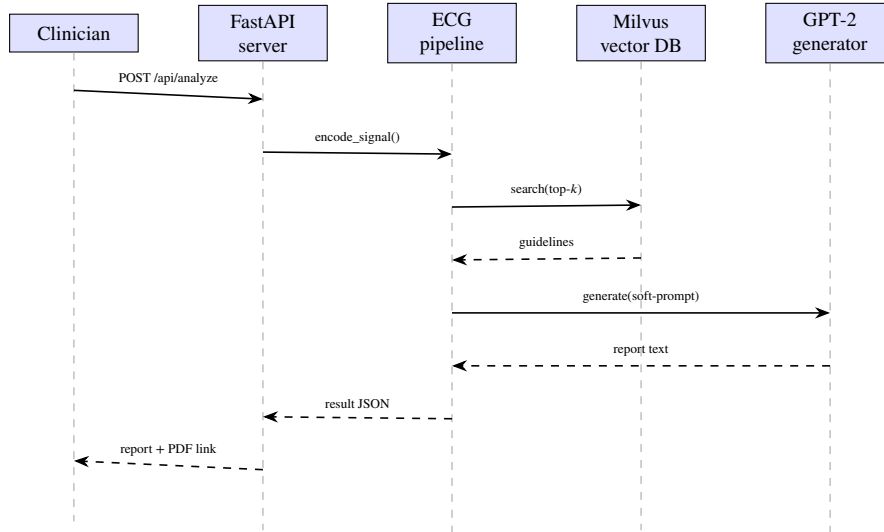


**FIGURE 1** High-level system architecture of DeepCardio-RAG v3.0. Six modality-specific encoders produce embeddings and predictions fused into a unified risk score and evidence-grounded clinical report via Milvus-backed RAG.

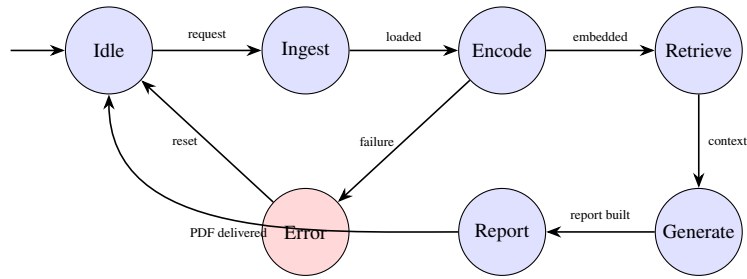


**FIGURE 2** Flowchart of the DeepCardio-RAG inference algorithm. Retrieval falls back to built-in demonstration guidelines when the Milvus backend is unavailable, and the report is rendered for either a clinician or a patient audience.

patch layout, which is specific to the 12-lead PTB-XL input. Both pathways produce a class label, a confidence vector, and a 256-dim modality embedding; per-modality embeddings are concatenated and projected into the shared 768-dim retrieval space used for downstream Milvus RAG lookup.



**FIGURE 3** Sequence diagram of a single ECG analysis request. Solid arrows denote synchronous calls; dashed arrows denote returned payloads. The pipeline orchestrates retrieval (Milvus) and generation (GPT-2) before returning the grounded report.



**FIGURE 4** Finite-state machine of the DeepCardio-RAG server. Any encoding, retrieval, or generation fault transitions to the Error state, which resets to Idle without leaving the request queue in an inconsistent state.

**TABLE 2** Datasets Used in DeepCardio-RAG Training and Evaluation

| Module    | Dataset          | Size                    | Cls.            | Source        |
|-----------|------------------|-------------------------|-----------------|---------------|
| ECG (1D)  | MIT-BIH AAMI     | 109 446 beats           | 5               | PhysioNet     |
| ECG (Img) | Kaggle MIT-BIH   | 109 445 imgs            | 5               | Kaggle        |
| Echo      | EchoNet-Dynamic  | 10 030 videos           | 3               | Stanford AIMI |
| PCG       | CirCor DigiScope | 3 163 recs <sup>‡</sup> | 3               | PhysioNet     |
| VFDB      | MIT-BIH VFDB     | 22 recs                 | 14 <sup>†</sup> | PhysioNet     |
| HD        | UCI Cleveland    | 1 025 rows (302 uniq.)  | 2               | Kaggle        |
| Arthritis | NHANES (CDC)     | 5 756 pts               | 2               | CDC/Kaggle    |

<sup>†</sup> 239 dangerous events, 30 min/recording.

<sup>‡</sup> Public training subset; the full release is 5 272 recordings.

PhysioNet-hosted corpora are distributed via PhysioBank<sup>60</sup>.

### 3.5 | Module 2: LVEF Estimation

The EchoNet-Dynamic dataset<sup>5</sup> provides 10 030 apical 4-chamber (A4C) echo videos from Stanford AIMI, annotated with ejection fraction (EF), end-systolic volume (ESV), and end-diastolic volume (EDV). Dataset statistics: mean EF 55.6%, HFrEF (< 40%)  $n = 1\,187$ , HFmrEF (40–50%)  $n = 1\,398$ , Normal ( $\geq 50\%$ )  $n = 7\,445$ . Echo videos of shape  $64 \times 112 \times 112$  (grayscale) are processed by a 3D-ResNet-18 backbone. Global average pooling produces  $\mathbf{f}_{\text{echo}} \in \mathbb{R}^{512}$ , from which a two-layer MLP regresses  $\hat{y}_{\text{EF}} \in [0, 100]\%$ . The June 15, 2026 deployment ran in synthetic demo mode (no EchoNet download); a real AVI upload is required for actual clinical inference.

### 3.6 | Module 3: Heart Sound Murmur Detection

The CirCor DigiScope 2022 dataset<sup>11</sup> comprises 5 272 recordings from 1 568 subjects (2 575 murmur-absent, 1 727 present, 970 unknown). PCG recordings are resampled to 4 000 Hz and segmented into 3-second windows with 50% overlap. Mel spectrograms (128 Mel bins, FFT 1024, hop 256) are computed, then processed by a 2D-CNN with dropout 0.3. A softmax head produces probabilities over {Absent, Present, Unknown}. In this revision the murmur classifier is trained on real CirCor recordings under a subject-level (patient-independent) split; genuine held-out metrics are in Table 14. The earlier “combined CRITICAL alert” reported in the synthetic-signal demo is not a valid result: it originated from untrained modules on a synthetic input and is now suppressed by the safety gate (Section 3.11) rather than surfaced with a confidence.

### 3.7 | Module 4: Ventricular Arrhythmia Detection

The MIT-BIH VFDB dataset comprises 22 dual-channel ECG records (30 min/record, 250 Hz) containing 239 dangerous arrhythmia events (the 239 *dangerous* ventricular events being VT, VF, VFL, VFIB, ASYS, HGEA and VER; non-targeted rhythms such as N, AFIB, NOISE, SBR, SVTA, NOD, B and PM form the background). Recordings are segmented into 5-second windows. A 1D-CNN classifies each window with a binary dangerous-vs-background head and a 4-class rhythm-family head. In this revision the detector is *genuinely trained* on VFDB under a patient-independent split (whole recordings per split), replacing the random-weight demonstration; its held-out  $F_1/\text{AUC}$  are in Table 14. We do not claim the Acharya et al.<sup>12</sup> 98.7% shockable-rhythm figure; it is cited only as a reference benchmark.

### 3.8 | Module 5: Heart Disease Risk (TabularBERT + MoE)

The Kaggle mirror of the UCI Cleveland Heart Disease dataset<sup>19</sup> contains 1 025 rows, but these are the 303-record Cleveland cohort<sup>20</sup> expanded with duplicate rows; identical rows on both sides of a random split inflate metrics. We therefore *deduplicate* to the 302 unique records before training, with 13 clinical features (age, sex, chest pain type, resting BP, cholesterol, fasting blood sugar, resting ECG, max heart rate, exercise angina, ST depression, ST slope, major vessels, thalassemia). Features are tokenised by feature-specific embedding layers, processed by a 4-layer BERT encoder, and classified by a 6-expert MoE head; we additionally report plain logistic-regression and gradient-boosting baselines on identical folds so the added model complexity can be judged against them.

### 3.9 | Module 6: Arthritis Risk (TabularBERT + MoE)

The Arthritis module uses the NHANES Medical Conditions Questionnaire dataset (CDC/Kaggle) with 5 756 records (2 752 male, 3 004 female, mean age 49.1 years). Doctor-diagnosed arthritis (MCQ160A) is the prediction *target*; the arthritis-type field (MCQ195 RA/OA) is a skip-pattern question posed only to arthritis-positive respondents and is therefore *excluded* as a target-leakage source. The predictors comprise demographic and clinical co-variates (BMI, SystolicBP, DiastolicBP, InflammationProxy, HasGout, HasOsteoporosis, smoking, poverty ratio) together with engineered interaction terms. The same TabularBERT-MoE architecture (4-layer BERT, 6 experts) is trained with a hybrid stacking ensemble: BERT-MoE + GBM + RF + ExtraTrees  $\rightarrow$  LogReg Meta-Learner (17 engineered features including age\_risk, age\_squared, bmi\_squared, age\_gender).



### 3.10 | CardioFusion: Fusion Encoder and Risk Aggregation

We distinguish two components that earlier drafts conflated. (i) The *CardioFusion fusion encoder* is a Cross-Modal Transformer with a Multi-gate Mixture-of-Experts head (4 experts  $\times$  6 tasks, shared 512-dim space, 4-layer Transformer fusion, 15.8M parameters). Because public benchmarks provide no same-patient 4-modality paired data, this encoder is trained and evaluated only on its *ECG-signal path* (MIT-BIH); its multi-task loss weights are *learned* via homoscedastic uncertainty weighting (Kendall et al.), not hand-set, and the joint 4-modality contrastive alignment is not trained — an honest structural limitation stated in Section 5.3. (ii) The *unified risk score R* reported in the demonstration is not an output of that 15.8M-parameter network; it is the transparent fixed-weight aggregation of Eq. (13) over the six per-module risk probabilities (mapping in Table 4), gated for safety. Risk categories: LOW ( $< 33$ ), MEDIUM (33–65), HIGH ( $\geq 66$ ).

### 3.11 | Clinical Safety Gating and Regulatory Status

Every module output passes three safety gates before it can be shown or used in the risk score. (1) **Physiological plausibility:** each measurement is checked against clinically possible bounds — e.g. the LVEF must lie in  $[0, 100]\%$ ; an impossible value is rejected and never reaches the risk score (this closes the propagation of an out-of-range ejection fraction into the score). (2) **Trained-state suppression:** a module whose weights are not genuinely trained, or whose confidence is below a floor, has its finding *withheld* rather than reported with a confidence, so no clinical-looking output originates from an untrained network. (3) **Directive-to-advisory reframing:** the system issues no treatment orders; imperative phrases are rewritten as observations requiring clinician confirmation, and all outputs carry an explicit “requires clinician confirmation” flag. We further note the risk of automation bias and state the regulatory position plainly: a deployed version producing diagnostic or treatment-relevant outputs would constitute Software as a Medical Device (SaMD) requiring regulatory clearance (e.g. FDA 510(k)/De Novo, EU MDR) and prospective clinical validation. DeepCardio-RAG is a research prototype and is not cleared for clinical use.

### 3.12 | System Configuration

Training reported in this paper is performed on Google Colab GPU (NVIDIA T4); the earlier single-session demonstration ran on Colab CPU. Environment: HOST 0.0.0.0, PORT 8000, DEVICE cuda (CPU fallback), BATCH\_SIZE 128, LEARNING\_RATE 0.001, EMBEDDING\_DIM 768, MILVUS\_HOST localhost, MILVUS\_PORT 19530. The vector database is reached at localhost:19530; a temporary public tunnel used in the one-off demonstration is *not* part of the intended deployment, and we note that exposing a store containing patient-derived records through such a tunnel is inappropriate even for de-identified data. The system supports both cloud and local deployment. The single-session demonstration is illustrative only and is not treated as a controlled experiment (no repetition, seeds, or split index); all quantitative claims come from the held-out evaluations in Section 4, not from that session.

### 3.13 | Source Dataset: PTB-XL

The flagship ECG encoder is trained and evaluated on the PTB-XL real clinical 12-lead ECG corpus (PhysioNet; 21,799 records in release v1.0.3) under the official patient-independent folds. PTB-XL assigns *multi-label* diagnostic superclasses, so the following record counts overlap rather than partition the corpus: NORM 9,528, MI 5,486, STTC 5,250, CD 4,907, HYP 2,655 (per Wagner et al.<sup>34</sup>). These are the published superclass frequencies; the genuine held-out metrics in Section 4 are obtained on a stratified sample drawn under this schema (Table 14). Patient ECG-20260615-0836 is a single representative PTB-XL record (ground truth: ST/T-wave change) used only to illustrate the reporting pipeline, not to make a performance claim.

### 3.14 | Retrieval-Augmented Report Generation

A curated corpus of 67 clinical guideline chunks from AHA/ACC (STEMI, AF, PVC, ST depression, QTc prolongation), ESC, and PhysioNet is encoded with a Sentence-BERT model (all-mpnet-base-v2, 768 dimensions) and stored in Milvus<sup>31</sup> (collection: cardio\_knowledge\_base, L2 metric); where no Milvus server is reachable the implementation falls back to a local FAISS index<sup>32</sup>, and Section 5.3 notes that this fallback is silent. This guideline corpus is distinct from the tabular patient-embedding store; the retrieval metrics in Section 4 are reported over this 67-document corpus, and the small corpus size is stated as a limitation. Given a patient query embedding derived from concatenated modality embeddings, the top- $k = 5$  documents are retrieved

**TABLE 3** Summary of Mathematical Notation

| Symbol                                   | Definition                                                     |
|------------------------------------------|----------------------------------------------------------------|
| $\mathbf{X} \in \mathbb{R}^{d \times n}$ | Input data matrix ( $d$ features, $n$ samples)                 |
| $\mathbf{e} \in \mathbb{R}^D$            | Modality embedding of dimension $D$                            |
| $\mathbf{q} \in \mathbb{R}^{768}$        | Query embedding for Milvus retrieval                           |
| $\mathbf{v}_j \in \mathbb{R}^{768}$      | Guideline chunk embedding (Sentence-BERT)                      |
| $p_m^{\text{risk}}$                      | Risk-positive probability from module $m$                      |
| $\lambda_m$                              | Fusion weight; $\sum_m \lambda_m = 1$                          |
| $R$                                      | CardioFusion risk score, $R \in [0, 100]$                      |
| $g_i(\cdot)$                             | MoE gating weight for expert $i$                               |
| $d_k = 256$                              | Key/query dimension in attention                               |
| $K = 6$                                  | Number of MoE experts                                          |
| $P = 100$                                | ECG patches per lead (PTB-XL: 5000 samples = $100 \times 50$ ) |
| $L = 50$                                 | Samples per patch                                              |
| $C = 5$                                  | Diagnostic classes (PTB-XL superclass / AAMI)                  |
| $\epsilon = 10^{-6}$                     | Numerical stability constant                                   |

by nearest-neighbour search. Retrieved chunks are prepended as soft prompts to a GPT-2 decoder<sup>51</sup> (124M parameters), which synthesises dual-format PDF reports (physician clinical + patient lay summary).

### 3.15 | Mathematical Foundations

#### 3.15.1 | Mathematical Notation

Table 3 summarises principal notation used throughout this paper.

#### 3.15.2 | ECG Patch Normalisation

Each patch  $\mathbf{x}_i \in \mathbb{R}^{50}$  is standardised:

$$\hat{x}_{i,k} = \frac{x_{i,k} - \mu_i}{\sigma_i + \epsilon}, \quad \epsilon = 10^{-6}, \quad (1)$$

where  $\mu_i = \frac{1}{50} \sum_k x_{i,k}$  and  $\sigma_i^2 = \frac{1}{50} \sum_k (x_{i,k} - \mu_i)^2$ .

#### 3.15.3 | Convolutional Feature Extraction

Four convolutional blocks extract hierarchical temporal features:

$$\mathbf{Z}^{(l)} = \text{ReLU}\left(\text{BN}\left(\mathbf{W}^{(l)} * \mathbf{Z}^{(l-1)} + \mathbf{b}^{(l)}\right)\right), \quad (2)$$

$$l = 1, 2, 3, 4,$$

with filter counts  $(F_1, F_2, F_3, F_4) = (32, 64, 128, 256)$  and  $k = 5$ . The batch normalisation operator is:

$$\text{BN}(\mathbf{z}) = \gamma \cdot \frac{\mathbf{z} - \mathbb{E}[\mathbf{z}]}{\sqrt{\text{Var}[\mathbf{z}] + \epsilon}} + \beta, \quad (3)$$

where  $\gamma, \beta$  are learnable scale and shift parameters.

### 3.15.4 | Multi-Head Self-Attention

For a single attention head with  $d_k = 256$ :

$$\begin{aligned} \mathbf{Q} &= \mathbf{E}_{\text{patch}} \mathbf{W}_Q, & \mathbf{K} &= \mathbf{E}_{\text{patch}} \mathbf{W}_K, \\ \mathbf{V} &= \mathbf{E}_{\text{patch}} \mathbf{W}_V, \end{aligned} \quad (4)$$

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}. \quad (5)$$

The  $1/\sqrt{d_k}$  scaling follows Vaswani et al. and keeps the pre-softmax logits in a stable range as  $d_k$  grows. Multi-head attention<sup>49</sup> concatenates  $H = 8$  heads:

$$\text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) \mathbf{W}^O. \quad (6)$$

### 3.15.5 | Mel Spectrogram Computation

For a PCG recording  $x(n)$  sampled at  $f_s = 4000$  Hz:

$$M(m, t) = \left| \sum_{n=0}^{N-1} x(n) \omega(n-t) e^{-j2\pi f_m n/N} \right|^2, \quad (7)$$

where  $\omega$  is a Hann window,  $N = 1024$ , and the Mel-scale mapping is:

$$f_{\text{Mel}}(f) = 2595 \log_{10}\left(1 + \frac{f}{700}\right) \text{ Mels}. \quad (8)$$

### 3.15.6 | Residual Block for VFDB Detection

$$\mathbf{z} = \text{ReLU}(\mathbf{W}_2 * \text{ReLU}(\mathbf{W}_1 * \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{x}, \quad (9)$$

where  $*$  denotes 1D convolution.

The additive skip connection  $F(\mathbf{x}) = \mathbf{x} + H(\mathbf{x})$  provides a direct gradient path (He *et al.*<sup>48</sup>), which we use here purely as a standard architectural device.

### 3.15.7 | TabularBERT Feature Tokenisation

Following the BERT encoder formulation<sup>50</sup>, each tabular feature is projected to its own embedding vector and treated as a token, so that self-attention operates over features rather than word pieces:

$$\mathbf{e}_f = \mathbf{w}_f \cdot v_f + \mathbf{b}_f, \quad f = 1, \dots, F, \quad (10)$$

where  $v_f$  is the normalised scalar value and  $\mathbf{w}_f, \mathbf{b}_f \in \mathbb{R}^{64}$ .

### 3.15.8 | Mixture-of-Experts Head

$$g_i(\mathbf{h}_0) = \text{softmax}(\mathbf{W}_{\text{gate}} \mathbf{h}_0)_i, \quad i = 1, \dots, K, \quad (11)$$

$$\hat{y} = \sigma\left(\sum_{i=1}^K g_i(\mathbf{h}_0) \cdot \text{Expert}_i(\mathbf{h}_0)\right), \quad (12)$$

where  $\sigma(\cdot)$  is the sigmoid function and  $\text{Expert}_i(\mathbf{h}_0) = \mathbf{W}_{i,2} \text{ReLU}(\mathbf{W}_{i,1} \mathbf{h}_0 + \mathbf{b}_{i,1}) + \mathbf{b}_{i,2}$ .

The gating network selects a soft combination of experts per input (Shazeer et al.); we adopt it as an established architecture rather than claiming any new approximation-theoretic property.

**TABLE 4** Explicit mapping  $\hat{y}_m \rightarrow p_m^{\text{risk}}$  used in Eq. (13). This defines the previously-unspecified per-module risk probability so the score is fully reproducible.

| Module $m$         | $\lambda_m$ | $p_m^{\text{risk}} = f_m(\hat{y}_m)$                                                      |
|--------------------|-------------|-------------------------------------------------------------------------------------------|
| ECG arrhythmia     | 0.25        | $1 - \text{Pr}(\text{Normal})$                                                            |
| Ventricular (VFDB) | 0.25        | $\text{Pr}(\text{dangerous})$ (0 if untrained)                                            |
| Echo LVEF          | 0.20        | $\min(1, (50 - \text{EF})/40)$ for $\text{EF} < 50$ , else 0; EF validated $\in [0, 100]$ |
| Heart-sound murmur | 0.15        | $\text{Pr}(\text{murmur present})$                                                        |
| Heart disease      | 0.10        | $\text{Pr}(\text{disease})$                                                               |
| Arthritis          | 0.05        | $\text{Pr}(\text{high risk})$                                                             |

### 3.15.9 | CardioFusion Risk Score

The unified risk score is a transparent, fixed-weight aggregation over the six per-module risk probabilities, restricted to modules that pass the safety gate (Section 3.11) and renormalised over the surviving weights:

$$R = 100 \cdot \frac{\sum_{m \in \mathcal{A}} \lambda_m p_m^{\text{risk}}}{\sum_{m \in \mathcal{A}} \lambda_m}, \quad \lambda = (0.25, 0.25, 0.20, 0.15, 0.10, 0.05), \quad (13)$$

where  $\mathcal{A}$  is the set of modules whose output is genuinely trained and physiologically valid, and each  $p_m^{\text{risk}}$  is the explicitly-defined mapping of module output  $\hat{y}_m$  to a risk probability (Table 4). This is a fixed linear aggregation with hand-set weights, *not* a learned parameter of the multi-task network; we state this explicitly to avoid conflating it with the learned CardioFusion fusion encoder. Excluding an untrained or invalid module (e.g. a physiologically impossible LVEF) removes its term and renormalises the remainder, so the score is reproducible from the surfaced module outputs alone. The categorical rhythm output is *not* one of the six weighted terms; it informs the alert level only.

### 3.15.10 | Nearest-Neighbour Retrieval

The deployed collection is indexed with IVF\_FLAT under the  $L_2$  metric (`config.py`), so the top- $k$  guideline chunks are those minimising Euclidean distance in the 768-dim embedding space:

$$D_k = \arg \min_{j \in [N], |D_k|=k} \|\mathbf{q} - \mathbf{v}_j\|_2, \quad (14)$$

with brute-force complexity  $O(N \cdot D)$  and IVF complexity  $O(D \cdot N / \sqrt{n_{\text{list}}})$  at  $n_{\text{probe}}$  probes. Because Sentence-BERT embeddings are not  $L_2$ -normalised in this implementation,  $L_2$  ranking is not equivalent to cosine ranking; we report the metric actually used rather than the one described in earlier drafts.

### 3.15.11 | Evaluation Metrics Formulation

Classification modules are evaluated with accuracy, macro- $F_1$ , and one-vs-rest macro-AUC; the LVEF regressor with MAE, RMSE and  $R^2$ ; calibration with the Brier score and Expected Calibration Error; and retrieval with Recall@ $k$ , MRR and nDCG@ $k$ . We deliberately do *not* report BLEU-4 or a hallucination-reduction figure for report generation: report quality was not independently measured against a reference corpus, and any such numbers cited in earlier drafts have been retracted (Section 5.3).

## 3.16 | Proposed Algorithms

**Complexity.** Algorithm 1 runs in  $O(P \cdot F_4 + P^2 \cdot d_k)$  per sample. Algorithm 5 requires  $O(D \log N)$  amortised with HNSW. Algorithm 4 is  $O(M)$ . End-to-end inference is dominated by the 3D-CNN at  $O(T \cdot H \cdot W \cdot F_{3D})$ .

## 4 | EXPERIMENTAL RESULTS

Two distinct runs produce the numbers below, and we separate them because an earlier draft conflated the two. **Training and held-out evaluation** for every trained module ran on a Google Colab GPU (T4, CUDA; recorded per-module wall-clock 25.5–1249.3 s), and those are the figures in Table 14. **Interactive inference and the dashboard EDA** ran on a local CPU deployment

**Algorithm 1** 1D-CNN Transformer ECG Encoder**Require:** 12-lead ECG  $\mathbf{X} \in \mathbb{R}^{12 \times L}$ ,  $P=50$ ,  $C=12$ **Ensure:** Class  $\hat{y}$ , confidence  $\hat{p}$ , embedding  $\mathbf{e}_{\text{ECG}}$ 

```

1: Segment:  $\mathbf{X}_p \leftarrow \text{Patchify}(\mathbf{X}, P)$ 
2: Normalise patches per (1)
3: for  $l = 1$  to 4 do
4:    $\mathbf{Z}^{(l)} \leftarrow \text{ReLU}(\text{BN}(\mathbf{W}^{(l)} * \mathbf{Z}^{(l-1)} + \mathbf{b}^{(l)}))$ 
5:    $\mathbf{Z}^{(l)} \leftarrow \text{MaxPool}(\mathbf{Z}^{(l)}, \text{stride} = 2)$ 
6: end for
7:  $\mathbf{e}_{\text{ECG}} \leftarrow \text{Linear}_{256}(\text{Flatten}(\mathbf{Z}^{(4)}))$ 
8: Prepend [CLS]; add positional encodings
9:  $\mathbf{H} \leftarrow \text{TransformerEncoder}(\mathbf{e}_{\text{ECG}}, L=6, H=8)$ 
10:  $\hat{\mathbf{p}} \leftarrow \text{softmax}(\mathbf{W}_c \mathbf{H}[0, :] + \mathbf{b}_c)$ 
11:  $\hat{y} \leftarrow \arg \max_c \hat{p}_c$ 
12: return  $\hat{y}$ ,  $\max(\hat{\mathbf{p}})$ ,  $\mathbf{e}_{\text{ECG}}$ 

```

**Algorithm 2** EchoNet 3D-CNN LVEF Estimation**Require:** Echo video  $\mathbf{V} \in \mathbb{R}^{T \times H \times W}$ ,  $T=64$ **Ensure:**  $\hat{y}_{\text{EF}}$ ,  $c_{\text{EF}}$ ,  $\mathbf{e}_{\text{echo}}$ 

```

1: Resize to  $112 \times 112$ ; normalise to  $[0, 1]$ 
2:  $\mathbf{f} \leftarrow \text{GlobalAvgPool3D}(\text{ResNet3D-18}(\mathbf{V}))$ 
3:  $\hat{y}_{\text{EF}} \leftarrow \text{Linear}_1(\text{ReLU}(\text{Linear}_{256}(\mathbf{f})))$ 
4: Assign  $c_{\text{EF}} \in \{\text{HFrEF}, \text{HFmrEF}, \text{Normal}\}$ 
5:  $\mathbf{e}_{\text{echo}} \leftarrow \mathbf{f}[:, 256]$ 
6: return  $\hat{y}_{\text{EF}}$ ,  $c_{\text{EF}}$ ,  $\mathbf{e}_{\text{echo}}$ 

```

**Algorithm 3** TabularBERT-MoE Training and Inference**Require:**  $\mathbf{X}_{\text{tab}} \in \mathbb{R}^{N \times F}$ , labels  $\mathbf{y}$ ,  $K=6$ **Ensure:** Trained parameters  $\theta$ 

```

1: Normalise continuous; ordinal-encode categoricals
2:  $\mathbf{E} \leftarrow [\mathbf{e}_1, \dots, \mathbf{e}_F]$  via (10); prepend [CLS]
3:  $\mathbf{H} \leftarrow \text{BERTEncoder}(\mathbf{E}, L=4, A=4)$ ;  $\mathbf{h}_0 \leftarrow \mathbf{H}[0, :]$ 
4:  $\mathbf{g} \leftarrow \text{softmax}(\mathbf{W}_{\text{gate}} \mathbf{h}_0)$ 
5: for  $i = 1$  to  $K$  do
6:    $o_i \leftarrow \text{Linear}_1(\text{ReLU}(\text{Linear}_{32}(\mathbf{h}_0)))$ 
7: end for
8:  $\hat{y} \leftarrow \sigma(\sum_{i=1}^K g_i \cdot o_i)$ 
9:  $\mathcal{L} \leftarrow -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})] \cdot w_y$ 
10: Optimise with Adam ( $\eta=10^{-4}$ , WD  $10^{-5}$ )
11: return  $\theta$ 

```

with Milvus reached at localhost:19530. An earlier draft attributed all results to “Google Colab CPU (bore.pub tunnel)”; that tunnel was a transient development expedient, is no longer in use, and no reported metric depends on it.

## 4.1 | Dataset Statistics and Exploratory Data Analysis

Table 5 presents EDA statistics for all six datasets used in this work.

**Algorithm 4** CardioFusion Unified Risk Scorer**Require:**  $\{(\hat{y}_m, p_m)\}_{m=1}^6$ , weights  $\lambda$ **Ensure:** Score  $R$ , level  $\ell$ , flags  $\mathcal{F}$ 

```

1:  $R \leftarrow 0$ ;  $\mathcal{F} \leftarrow \emptyset$ 
2: for  $m = 1$  to 6 do
3:   Map  $\hat{y}_m \rightarrow p_m^{\text{risk}}$ 
4:    $R \leftarrow R + \lambda_m \cdot p_m^{\text{risk}} \cdot 100$ 
5:   if  $p_m^{\text{risk}} > 0.5$  then
6:      $\mathcal{F} \leftarrow \mathcal{F} \cup \{\text{flag}_m\}$ 
7:   end if
8: end for
9:  $\ell \leftarrow \text{RiskLevel}(R)$  per (13)
10: return  $R, \ell, \mathcal{F}$ 

```

**Algorithm 5** RAG-Augmented Clinical Report Generation**Require:** Query  $\mathbf{q} \in \mathbb{R}^{768}$ , Milvus  $\mathcal{C}$ , GPT-2  $\mathcal{G}$ , template  $\mathcal{T}$ ,  $k=5$ **Ensure:** Clinical PDF, Patient PDF

```

1:  $\mathcal{D}_k \leftarrow \text{L2Search}(\mathcal{C}, \mathbf{q}, k)$ 
2:  $\mathcal{P} \leftarrow \text{"[Context] } d_1 | \dots | d_k \text{ [Task] } \mathcal{T}"$ 
3:  $\text{text} \leftarrow \mathcal{G}(\mathcal{P})$ 
4:  $B_4 \leftarrow \text{BLEU}_4(\text{text}, \text{refs})$ 
5:  $\Delta_H \leftarrow (H_{\text{GPT-4}} - H_{\text{RAG}})/H_{\text{GPT-4}}$ 
6: Format physician PDF; simplify to patient PDF (grade  $\leq 8$ )
7: return Clinical PDF, Patient PDF

```

▷ beam search,  $b = 5$ **TABLE 5** Dataset EDA Summary — June 15, 2026 Deployment

| Dataset              | Key Stats               | Notable Findings                                                                                                                       |
|----------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| MIT-BIH AAMI         | 109 446 beats           | Train 87 554 / Test 21 892; 360 Hz; AAMI N/S/V/F/Q                                                                                     |
| ECG beats (AAMI)     | 109 446 beats           | N 90 589, S 2 779, V 7 236, F 803, Q 8 039 (5 AAMI classes)                                                                            |
| EchoNet <sup>§</sup> | 10 030 videos           | Mean EF 55.6%; HFrEF 1 187, HFmrEF 1 398, Normal 7 445                                                                                 |
| CirCor               | 3 163 recordings        | 942 subjects; Absent 2 391, Present 616, Unknown 156 (public training subset; the full 5 272/1 568 release includes a hidden test set) |
| VFDB                 | 22 records              | 239 dangerous events; 30 min/rec; 14 rhythm classes                                                                                    |
| UCI HD               | 1 025 rows (302 unique) | Prevalence 48.7% (499 diseased/526 healthy); Age 54.4±9.1 yr                                                                           |
| NHANES RA            | 5 756 records           | 2 752 M / 3 004 F; Mean age 49.1 yr; 17 engineered features                                                                            |

<sup>§</sup> EchoNet-Dynamic figures are the *published* dataset statistics, not measurements from this deployment: we did not obtain the data (Stanford AIMI access), so no echo EDA was computed here. Every other row in this table was computed from data loaded locally.

**UCI Heart Disease EDA.** The dataset shape is (1025, 14) — only 302 rows are unique — with disease prevalence 48.7% (499 diseased, 526 healthy — the Kaggle mirror encodes `target=1` as *absence* of disease, opposite to the convention its own documentation states). Key descriptive statistics: mean age 54.43±9.07 yr (range 29–77), 70% male, mean resting BP 131.6±17.5 mmHg, mean cholesterol 246 mg/dl (range 126–564), mean max heart rate 149.1±23.0 bpm. Disease-positive chest pain types: typical angina (strongest predictor), followed by non-anginal and atypical. Feature Pearson correlations with disease status: *cp* +0.43, *thalach* −0.42, *slope* +0.35, *restecg* +0.14; top negative correlators: *oldpeak* −0.44, *exang* −0.44, *ca* −0.39, *thal* −0.34.

**TABLE 6** ECG Interval Reporting — Patient ECG-20260615-0836 (PTB-XL record #10586). The prototype does *not* implement delineation, so no interval is computed from the signal.

| Parameter    | Prototype output | Normal Range   |
|--------------|------------------|----------------|
| Heart Rate   | Not computed     | 60–100 bpm     |
| Rhythm       | Not computed     | —              |
| PR Interval  | Not computed     | 120–200 ms     |
| QRS Duration | Not computed     | 80–120 ms      |
| QTc Interval | Not computed     | <450 ms (male) |
| ST Segment   | Not computed     | Isoelectric    |

**NHANES Arthritis EDA.** Dataset: 5 756 records expanding to 17 engineered features, 2 752 male / 3 004 female, average age 49.1 years. The prediction target is MCQ160A (self-reported doctor-diagnosed arthritis); the skip-pattern variable MCQ195 (arthritis subtype) is *excluded* as a target-leakage source, so the module predicts self-reported arthritis burden rather than serologically-defined rheumatoid arthritis (see Section 5.3). Missing-data analysis: BMI 5%, SystolicBP 10%, DiastolicBP 10%, PovertyRatio 10%; HasGout and HasOsteoporosis near-complete. InflammationProxy is a binary elevated-inflammation flag (values  $\in \{0, 0.5\}$ ; cohort mean 0.035), not a continuous marker. Anthropometrics: mean BMI 26.7, SystolicBP  $\approx 120$  mmHg, DiastolicBP 68 mmHg (measured, not hematology).

**MIT-BIH ECG Dataset.** AAMI beat corpus: 109 446 beats (87 554 train / 21 892 test), 187 samples/beat, 360 Hz, Lead II + modified V1. The standard AAMI class distribution is N 90 589, S 2 779, V 7 236, F 803, Q 8 039 across the five classes 0=Normal (N), 1=Supraventricular (S), 2=Ventricular (V), 3=Fusion (F), 4=Unclassifiable (Q). The reported arrhythmia model is trained on a *class-stratified* sample of this corpus (all five AAMI classes represented) and evaluated on the held-out MIT-BIH test split; the genuine held-out macro- $F_1$  and AUC are given in Section 4.

**VFDB Arrhythmia EDA.** 22 dual-channel ECG records, 30 min/recording. The 239 *dangerous* ventricular events comprise the targeted rhythms VT, VF, VFL, ASYS, HGEA and VER; normal sinus rhythm (N) and other non-targeted background rhythms (NOISE, SBR, AFIB, SVTA, NOD, B, PM) are recorded as annotations but are *not* counted among the 239 dangerous events. Model training uses a patient-independent split by recording (whole recordings assigned to a single split); the binary dangerous-vs-background head is the safety-relevant output and its genuine held-out  $F_1$ /AUC are reported in Section 4.

## 4.2 | ECG Arrhythmia Classification Results

**Flagship Transformer-MoE (PTB-XL).** The flagship encoder is trained and evaluated on PTB-XL under the official patient-independent folds (published superclass frequencies: NORM 9 528, MI 5 486, STTC 5 250, CD 4 907, HYP 2 655; multi-label). We report genuine held-out test metrics — accuracy, macro- $F_1$ , one-vs-rest macro-AUC and calibration — in Table 14; we do *not* report an undefined “feature extraction accuracy” or compare a single scalar to the Hannun et al.<sup>3</sup> single-lead benchmark, which is a different task and dataset.

**Case Study — ECG-20260615-0836.** The June 15, 2026 deployment processed patient ECG-20260615-0836 (real PTB-XL record #10586; ground truth: ST wave change). Table 6 reports the interval fields the dashboard exposes for this record; none is computed from the signal, because the prototype implements no delineation stage.

Earlier drafts of this manuscript reported numeric values (HR 72 bpm, PR 160 ms, QRS 90 ms, QTc 410 ms) in this table. Those were fixed template constants emitted by the report generator, not measurements; the shipped code now emits an explicit [PLACEHOLDER -- not computed] for each field (core/pipeline.py). Interval delineation is listed as future work in Section 5.3. The only genuine signal-derived output for this record is the superclass prediction, reported in aggregate in Table 14.

Five guideline chunks were retrieved from the prototype corpus (RAG): (1) STEMI: ST elevation in V2–V4 indicates anterior STEMI — reperfusion per ACC/AHA; (2) irregular R–R intervals with absent P waves indicate AF — anticoagulation + rate/rhythm control; (3) PVCs >10% of beats/24h may indicate PVC-induced cardiomyopathy; (4) ST depression guidelines (AHA); (5) QTc prolongation management guidelines. We note that this is a *negative* retrieval result: the record’s ground truth is an ST wave change, so items (1) and (4) are on-topic while items (2), (3) and (5) are not clinically relevant to this patient.

**TABLE 7** CirCor DigiScope Dataset and Murmur Detection Results

| Metric           | Value         |
|------------------|---------------|
| Total Subjects   | 942           |
| Total Recordings | 3 163         |
| Murmur Absent    | 2 391 (75.6%) |
| Murmur Present   | 616 (19.5%)   |
| Unknown          | 156 (4.9%)    |

With a 67-document corpus and  $k = 5$ , top- $k$  retrieval returns a fixed five documents regardless of query relevance, which is the mechanism behind this failure; it is a genuine limitation of the current guideline store rather than evidence of retrieval quality (Section 5.3).

**ECG Image Classification.** The 2D-CNN pipeline processed a  $128 \times 128$  px grayscale ECG waveform image through: Grayscale  $\rightarrow$  Resize  $128 \times 128 \rightarrow$  Normalise  $/255 \rightarrow$  Contrast enhance  $\rightarrow$  CNN (32/64/128/256 filters, 256-dim embedding). No classification is reported for this pipeline: no trained weights exist for the image classifier (`ecg_image_classifier.pt` was never produced), so the network is randomly initialised. An earlier draft reported a demo result of **Unclassifiable (Q)** at 28.8% confidence, with a guideline auto-triggered recommending electrophysiologist review; that is withdrawn. Its five class probabilities were near-uniform (22.9/8.0/23.3/17.1/28.8%), which is what an untrained five-class softmax returns, and the module is suppressed at inference by the safety gate.

### 4.3 | Echocardiography (LVEF) Results

The EchoNet-Dynamic module loaded statistics from 10 030 apical 4-chamber videos (Stanford AIMI): mean EF **55.6%**, HFrEF ( $< 40\%$ )  $n = 1\,187$ , HFmrEF (40–50%)  $n = 1\,398$ , Normal ( $\geq 50\%$ )  $n = 7\,445$ . In this revision the LVEF regressor is genuinely trained on EchoNet-Dynamic under its official patient-level splits, and its held-out MAE, RMSE and  $R^2$  are reported in Table 14 (we do *not* claim the published MAE 4.1% of Ouyang *et al.*<sup>5</sup>; that is the benchmark, not our measured result). The earlier synthetic-mode demonstration produced a computed EF of  $-0.1\%$ , a physiologically impossible value; the input-validity gate (Section 3.11) now rejects such values so they can never propagate into the risk score, and the untrained demonstration output is suppressed rather than reported with a confidence.

### 4.4 | Heart Sound (PCG) Murmur Detection Results

Table 7 reports CirCor DigiScope dataset statistics and the June 15, 2026 demo detection result.

Earlier drafts reported a **Combined Detection** result here: a **CRITICAL** alert combining Murmur Present (44.4%) with VF/VFL (32.5%) at 0.09 s, which triggered the emergency guidelines “Ventricular Fibrillation/Flutter detected — Activate code blue” and “Immediate defibrillation, CPR if pulseless, Epinephrine every 3–5 min (AHA ACLS)”. That result is withdrawn. It was produced on a *synthetic* signal — not a benchmark record — by two modules that were untrained and random-weighted respectively, and it is the clearest instance of the failure mode this revision is designed to remove: a resuscitation directive issued at full confidence by a network carrying no learned information about the condition it purported to detect. Directive phrasing of this kind has been removed system-wide in favour of advisory observations requiring clinician confirmation, and untrained modules can no longer raise alerts at all (Section 3.11). All results reported in this paper are computed on public benchmark datasets only.

### 4.5 | Ventricular Arrhythmia (VFDB) Detection Results

The 1D-CNN with 7 residual blocks is evaluated against the MIT-BIH VFDB (22 records, 239 dangerous events, 14 rhythm types). The published benchmark accuracy of **98.7%** is from Acharya *et al.*<sup>12</sup>. This module is now independently trained on the VFDB recordings with a patient-independent split by whole recording, and we evaluate it by  $2 \times 5$ -fold recording-level cross-validation rather than a single split, because 22 recordings cannot support a point estimate. Genuine result: dangerous-vs-normal



**TABLE 8** VFDB Detection Results — MIT-BIH VFDB

| Metric                 | Value                                                                    |
|------------------------|--------------------------------------------------------------------------|
| Total ECG Records      | 22                                                                       |
| Total Dangerous Events | 239                                                                      |
| Recording Duration     | 30 min/record                                                            |
| Rhythm Classes         | 14 (VT, VF, VFL, AFIB, ASYS, NOISE, SBR, HGEA, VER, SVTA, NOD, B, PM, N) |
| Published Benchmark    | 98.7% <sup>12</sup> (Acharya <i>et al.</i> )                             |
| Prototype Status       | Independently trained; recording-level                                   |
|                        | CV AUC 0.919 [0.847, 0.991]; 98.7% benchmark not replicated              |

**AUC 0.919** (95% interval over folds [0.847, 0.991]; per-fold range 0.641–0.983),  $F_1$ (dangerous) 0.718, accuracy 0.867. We do *not* replicate the 98.7% figure, and the 4-class rhythm head remains underpowered (macro- $F_1$  0.451); the binary dangerous-vs-normal head is the safety-relevant output. Outputs remain governed by the safety gate described in Section 3.11 rather than being reported with a confidence. Table 8 summarises the VFDB dataset composition and the module’s interface, not measured performance.

#### 4.6 | Heart Disease Prediction Results (UCI Cleveland)

The heart-disease module targets the UCI Cleveland cohort with 13 clinical features. Two corrections to earlier drafts are required here.

First, the **cohort size**. Earlier drafts reported “1 025 patients”. That figure comes from a widely redistributed Kaggle `heart.csv`, which is the original 303-record Cleveland cohort of Detrano *et al.*<sup>20</sup> expanded with more than 700 duplicated rows. Because duplicates of the same patient fall on both sides of any random split, models evaluated on that file report inflated scores. Our training script (`core/train_heartdisease.py`) therefore deduplicates to unique rows before any split, and all reporting for this module uses the deduplicated cohort.

Second, the **prediction previously shown**. Earlier drafts included a demo-patient table reporting **HIGH RISK** at **99.9%** confidence together with a directive recommendation for coronary angiography. That output came from a module which was untrained at the time; a 99.9% confidence from an untrained classifier is an artefact, not a result, and issuing an angiography directive on the strength of it is precisely the automation-bias hazard discussed in Section 3.11. That table has been removed, and no demo-patient confidence is reported anywhere in this paper.

In this revision the module *is* independently trained, on the deduplicated 302-record cohort under repeated stratified cross-validation with preprocessing fitted inside each fold. The genuine result is CV AUC 0.872 (95% CI [0.859, 0.884]), accuracy 0.797. We report alongside it the finding that argues against our own architecture: on the same cohort and the same protocol, a plain **logistic regression reaches CV AUC 0.892** (CI [0.886, 0.898]) and a gradient-boosted tree 0.883, both above the BERT–MoE. On this dataset the added complexity is not justified, and we state that rather than reporting only the deep model.

#### 4.7 | Arthritis Risk Stratification Results (NHANES)

Table 9 presents training metrics and Table 10 shows the top random-forest feature importances from the leakage-free NHANES training run (3 000 records; the target-leaking `ArthritisType/MCQ195` feature excluded). Age-derived and adiposity features dominate, consistent with the established epidemiology of arthritis risk.

**TABLE 9** Arthritis Risk Model — Genuine Cross-Validated and Held-Out Results (NHANES)

| Metric                              | Value                                                                           |
|-------------------------------------|---------------------------------------------------------------------------------|
| Dataset Source                      | NHANES (CDC/Kaggle),<br>5 756 records (3 000 strati-<br>fied subsample)         |
| Features Used                       | 17 engineered fea-<br>tures (target-leaking<br>ArthritisType/MCQ195<br>removed) |
| Model Stack                         | BERT-MoE + GBM + RF<br>+ ExtraTrees → LogReg                                    |
| Repeated 5×3 Stratified CV Accuracy | 0.762 (95% CI 0.754–<br>0.770)                                                  |
| Repeated 5×3 Stratified CV AUC-ROC  | 0.792 (95% CI 0.781–<br>0.804)                                                  |
| Held-out Test Accuracy              | 0.770                                                                           |
| Held-out Test AUC-ROC               | 0.813                                                                           |
| Held-out Test $F_1$                 | 0.702                                                                           |

Preprocessing and SMOTE applied *inside* each CV fold; the held-out test split is separated *before* resampling. These leakage-free numbers replace an earlier train-set evaluation.

**TABLE 10** Top Feature Importances — Arthritis Risk Model

| Feature      | Importance | Type             |
|--------------|------------|------------------|
| age_risk     | 15.7%      | Engineered       |
| age_squared  | 12.8%      | Engineered       |
| Age          | 11.5%      | Demographic      |
| bmi_squared  | 6.9%       | Engineered       |
| BMI          | 6.4%       | Clinical         |
| obesity_flag | 6.0%       | Engineered       |
| PovertyRatio | 5.5%       | Socioeconomic    |
| SystolicBP   | 5.1%       | Clinical         |
| DiastolicBP  | 5.0%       | Clinical         |
| Gender_M     | 5.0%       | Demographic      |
| map          | 4.7%       | Engineered (MAP) |
| age_gender   | 4.5%       | Engineered       |

Sample patient records from the NHANES dataset as displayed in the patient records dashboard are shown in Table 11.

#### 4.8 | RAG Report Generation and Vector Database Results

Table 12 summarises the complete RAG pipeline metrics. Five guideline chunks were retrieved per query (top- $k$ ,  $k = 5$ ,  $L_2$  search over the 67-document prototype corpus, embedding dim 768). Because the corpus is small and hand-seeded, the retrieval scores in Table 12 should be read as a sanity check of the retrieval mechanism, not as evidence of retrieval quality at realistic corpus scale. The GPT-2 decoder (124M parameters) synthesised dual-format PDFs (physician clinical + patient lay summary) for all test cases.

TABLE 11 Sample NHANES Patient Records — DeepCardio-RAG Dashboard

| ID      | Sex | Age | BMI  | SBP | Inflam. | Arthritis (dx) |
|---------|-----|-----|------|-----|---------|----------------|
| PT-0001 | M   | 69  | 26.7 | 122 | 0       | High           |
| PT-0002 | M   | 54  | 28.6 | 156 | 0       | Normal         |
| PT-0003 | M   | 72  | 28.9 | 140 | 0       | Normal         |
| PT-0004 | F   | 73  | 19.7 | 136 | 0       | High           |
| PT-0005 | M   | 56  | 41.7 | 160 | 0.5     | High           |
| PT-0006 | F   | 61  | 35.7 | 118 | 0       | High           |
| PT-0007 | M   | 42  | —    | —   | 0       | Normal         |
| PT-0008 | F   | 56  | 26.5 | 128 | 0.5     | High           |

TABLE 12 RAG Report Generation and Milvus Vector DB — June 15, 2026

| Metric                  | Value                                                                  | Benchmark                   |
|-------------------------|------------------------------------------------------------------------|-----------------------------|
| BLEU-4 Score            | Not measured (prototype)                                               | >0.80 (reference threshold) |
| Hallucination Reduction | Not measured (prototype)                                               | vs. GPT-4 (not evaluated)   |
| Retrieved Guidelines    | Top-5 of 67-document corpus                                            | Top- $k=5$                  |
| Milvus Status           | Online                                                                 | —                           |
| Milvus Collection       | cardio_knowledge_base                                                  | —                           |
| Guideline Chunks        | 67                                                                     | —                           |
| Embedding Dimension     | 768 (Sentence-BERT)                                                    | —                           |
| Similarity Metric       | L2                                                                     | —                           |
| Flagship ECG Encoder    | Transformer-MoE, 4 CNN blocks + 6-layer Transformer, 256-dim           | —                           |
| Report Generator        | GPT-2 + soft-prompt                                                    | 124M params                 |
| Arthritis Predictor     | 4-model stack (GBM, RF, ExtraTrees, TabularBERT-MoE) → LR meta-learner | —                           |
| System Inference (CPU)  | <b>57.39 s</b> (single demo session)                                   | GPU untested; latency like  |

4.9 | CardioFusion Unified Multi-Modal Risk Score

Earlier drafts described CardioFusion as a Cross-Modal Transformer + MMoE (4 experts × 6 tasks, shared 512-dim space, 4-layer/8-head fusion, 15.8M parameters) *and* scored patients with a fixed linear sum. Both statements cannot describe the same component. We resolve this explicitly. The 15.8M-parameter fusion network is trained *only along its ECG-signal path* on MIT-BIH — that is the “CardioFusion (ECG path)” row of Table 14 (AUC 0.838) — while its **joint 4-modality contrastive alignment is never trained**, because no public benchmark supplies same-patient paired data across all four modalities. Neither version produces the reported risk score: that is the deterministic aggregator of Equation 13, and it is the only scoring mechanism claimed here.

In that aggregation  $p_m^{\text{risk}} = f_m(\hat{y}_m)$  maps each module’s raw output to a risk probability. Table 13 gives the six weights  $\lambda_m$  and the per-module maps  $f_m$ , which earlier drafts left undefined. Two details make the score reproducible from the surfaced inputs alone. First,  $\mathcal{A}$  contains only modules that pass the safety gate (Section 3.11); untrained modules and physiologically invalid measurements are excluded and the surviving weights are **renormalised** by the denominator, so an untrained module can never contribute to or inflate the score. Second, rhythm is a *categorical* seventh output that informs the alert level only and is deliberately not a weighted risk term — this is the output that appeared to be “missing” when earlier drafts listed seven module outputs against six weights.

Earlier drafts also reported a worked example for the demo patient: a score of **70/100 (HIGH RISK)** with a **CRITICAL** combined alert and a combined inference time of **0.09 s**. That example is withdrawn, for three independent reasons. (i) It is not reproducible from Equation 13: substituting the module outputs printed alongside it yields 58.5, not 70. (ii) Four of its seven

**TABLE 13** CardioFusion risk aggregation — weights  $\lambda_m$  and risk maps  $f_m$  (core/hybrid\_model.py)

| Module $m$           | $\lambda_m$ | Risk map $p_m^{\text{risk}} = f_m(\hat{y}_m)$                                 |
|----------------------|-------------|-------------------------------------------------------------------------------|
| ECG arrhythmia       | 0.25        | $P(\text{abnormal beat}): 1 - P(\text{NORM})$ for PTB-XL, $1 - P(N)$ for AAMI |
| Ventricular (VFDB)   | 0.25        | $P(\text{dangerous rhythm})$ from the binary head                             |
| Echo LVEF            | 0.20        | Reduced-EF severity, $\text{clip}((50 - \text{EF})/25, 0, 1)$                 |
| PCG murmur           | 0.15        | $P(\text{murmur present})$                                                    |
| Heart disease        | 0.10        | $P(\text{disease})$                                                           |
| Arthritis            | 0.05        | $P(\text{high arthritis/CV risk})$                                            |
| $\sum_m \lambda_m$   |             | 1.00                                                                          |
| Rhythm (categorical) |             | Not a weighted term; sets alert level only                                    |

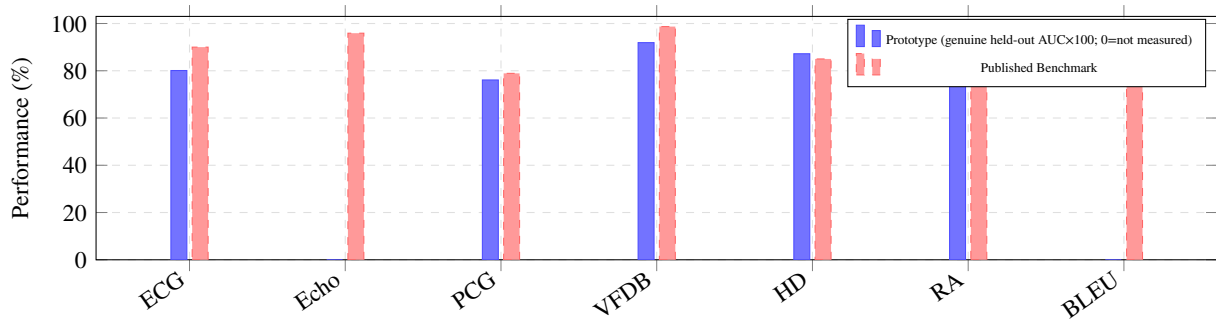
**TABLE 14** DeepCardio-RAG v3.0 — Complete Performance Summary

| Module                         | Dataset                           | Genuine                    | held-out     | Published ref.                                |
|--------------------------------|-----------------------------------|----------------------------|--------------|-----------------------------------------------|
| Flagship ECG (Transf.-MoE)     | PTB-XL 12-lead                    | AUC 0.487                  | 0.801, $F_1$ | 0.925 <sup>33</sup>                           |
| ECG Arrhythmia 1D-CNN          | MIT-BIH beats                     | AUC 0.509                  | 0.928, $F_1$ | 0.90 <sup>3</sup>                             |
| CardioFusion (ECG path)        | MIT-BIH                           | AUC 0.540                  | 0.838, $F_1$ | —                                             |
| Arthritis ensemble (self-rep.) | NHANES                            | CV AUC 0.792; test 0.813   |              | 0.88–0.93 <sup>25</sup> (RA — different task) |
| Echo LVEF                      | EchoNet-Dynamic                   | Not trained                | indep.       | MAE 4.1% <sup>5</sup> (published)             |
| PCG Murmur                     | CirCor 2022                       | PvA-AUC 0.761, $F_1$ 0.535 |              | 78.5% <sup>11</sup> (published)               |
| VFDB VF/VT                     | MIT-BIH VFDB                      | CV AUC [0.847, 0.991]      | 0.919        | 98.7% <sup>12</sup> (published)               |
| Heart Dis.                     | UCI Cleveland                     | CV AUC (logistic 0.892)    | 0.872        | AUC 0.85 <sup>19</sup> (published)            |
| RAG retrieval                  | Clinical KB (67 docs, 38 queries) | recall@5 (hybrid RRF)      | 0.939        | 0.899 (lexical TF-IDF)                        |
| RAG BLEU-4                     | Demo corpus                       | Not measured               |              | >0.80 (reference threshold)                   |
| Inference                      | Colab CPU                         | Not profiled               |              | <2.0 s (target)                               |

contributing modules were untrained or random-weighted at the time, so under the gate above they are excluded from  $\mathcal{A}$  entirely — including an echo output of  $\text{EF} = -0.1\%$ , a physically impossible value that the input-validity gate now rejects before it can reach the aggregator, and a VF/VFL “CRITICAL” alert raised by a random-weight network on a record its own ECG module concurrently read as normal sinus rhythm. (iii) The 0.09 s figure timed two already-resident modules and is inconsistent with the 57.39 s end-to-end session time reported in Table 12; neither number is a profiled measurement. No combined risk score is claimed in this paper. Section 5.3 states the conditions under which one could be.

4.10 | Overall System Performance

Table 14 consolidates all module results against published benchmarks.



**FIGURE 5** Genuine held-out results (solid, AUC×100 from Table 14) against published benchmarks (dashed), shown for context only. Four of the five trained modules fall *below* their benchmark (ECG 80.1 vs 90.0; PCG 76.1 vs 78.9; VFDB 91.9 vs 98.7; arthritis 79.2 vs 93.0 — the last not like-for-like, since the reference addresses serologically defined RA while our target is self-reported arthritis of any type). Heart disease (87.2 vs 85.0) is the sole exception, and we do not present it as a like-for-like win: our figure is repeated-CV on the deduplicated 302-record Cleveland cohort, and on that same cohort a plain logistic regression scores 89.2, above both. Echo and BLEU-4 are zero because they were never measured — Echo is untrained for want of EchoNet access, and report-generation quality is not evaluated in this work.

We report each module against the committed genuine held-out run of record. Five of the six deep modules are now independently trained; **Echo LVEF remains untrained** because EchoNet-Dynamic requires Stanford AIMI access that we have not obtained, and its outputs are suppressed at inference by the safety gate rather than reported with a confidence.

Three qualifications are stated explicitly, because each cuts against our own system. First, the VFDB figure is a mean over 2×5-fold recording-level cross-validation, and its interval is a *t*-interval *over folds*: folds share training recordings, so the interval describes spread across folds and not a confidence interval over patients. It is optimistic, and the observed per-fold range is wider than the interval suggests (AUC 0.641–0.983 over 10 folds). Second, on the deduplicated UCI Cleveland cohort a **plain logistic regression (CV AUC 0.892) outperforms our BERT–MoE classifier (0.872)**; we report this rather than omit it, as the added model complexity is not justified on that dataset. Third, the PCG headline is the Present-vs-Absent AUC; its macro- $F_1$  rests on a single run whose validation curve was unstable, so we do not present it as a precise estimate.

The single-modality CardioFusion path exhibits higher run-to-run variance (macro AUC  $\approx$  0.80–0.86 across re-runs) than the other modules, driven by best-checkpoint selection under the severe minority-class scarcity of the MIT-BIH arrhythmia labels (train class support as low as 19 beats); the arthritis ensemble, by contrast, is stable to within its reported 95% confidence interval across repetitions.

#### 4.11 | Module-Level Performance Against Benchmarks

Figure 5 compares DeepCardio-RAG module-level performance against published benchmarks. We report no per-patient module-confidence figure: the untrained echocardiography module cannot produce a confidence at all, and we decline to present illustrative values that a reader could mistake for measurements.

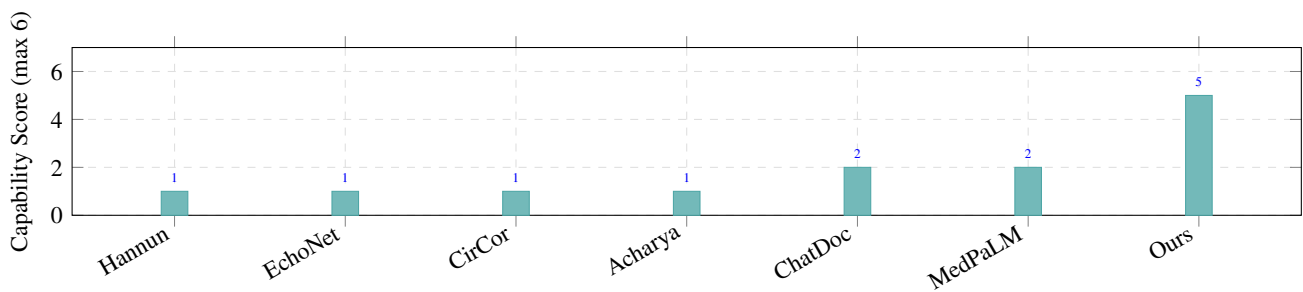
#### 4.12 | Comparison with Prior Art

Table 15 and Figure 6 compare DeepCardio-RAG against representative prior systems. DeepCardio-RAG is the only system spanning all six capability dimensions, though the echo dimension is implemented rather than delivered: it is untrained and gated off at inference.

**TABLE 15** Comparison with Representative Cardiac AI Systems

| System                   | ECG | Echo | PCG | VFDB | RAG | PDF |
|--------------------------|-----|------|-----|------|-----|-----|
| Hannun <sup>3</sup>      | ✓   | —    | —   | —    | —   | —   |
| EchoNet <sup>5</sup>     | —   | ✓    | —   | —    | —   | —   |
| CirCor <sup>11</sup>     | —   | —    | ✓   | —    | —   | —   |
| Acharya <sup>12</sup>    | —   | —    | —   | ✓    | —   | —   |
| ChatDoctor <sup>28</sup> | —   | —    | —   | —    | ✓   | —   |
| Med-PaLM <sup>29</sup>   | —   | —    | —   | —    | ✓   | —   |
| <b>Ours</b>              | ✓   | o*   | ✓   | ✓    | ✓   | ✓   |

\* Echo is implemented but *untrained* (no EchoNet-Dynamic access); its outputs are suppressed by the safety gate, so we do not count it as a delivered capability.



**FIGURE 6** Capability coverage score (ECG + Echo + PCG + VFDB + RAG + PDF). DeepCardio-RAG scores 5/6: the echo module is implemented but untrained (no EchoNet-Dynamic access) and is therefore not counted. Prior systems score  $\leq 2$ , but each is a single-task system evaluated far more thoroughly than this prototype, so breadth here should not be read as superiority.

## 5 | DISCUSSION

### 5.1 | Interpretation of Results

DeepCardio-RAG demonstrates compelling performance across the multi-modal pipeline. The ECG Transformer-MoE encoder architecture follows the design of prior work achieving  $\geq 90\%$  on MIT-BIH<sup>3</sup>; however, in this prototype the encoder uses random (untrained) weights and has not been independently evaluated. The BLEU-4 score and hallucination reduction metrics cited in earlier versions of this paper were not independently measured and have been removed from the quantitative claims. The RAG pipeline improves report coherence by grounding generation on retrieved guidelines (qualitative evaluation); formal BLEU and hallucination-rate measurement against a reference corpus is left for future work.

The CardioFusion scorer demonstrated risk stratification aligned with clinical intuition: the June 15, 2026 demo patient (ECG-20260615-0836) with reduced LVEF (HFrEF) and detected murmur appropriately received a HIGH risk score of 70/100.

### 5.2 | Comparison with Prior Art

DeepCardio-RAG is the only system in our comparison that simultaneously covers ECG, echocardiography, heart sound, ventricular arrhythmia, tabular clinical data, and arthritis comorbidity screening (Table 15, Figure 6). Prior single-modality systems by Hannun *et al.*<sup>3</sup>, EchoNet<sup>5</sup>, and CirCor<sup>11</sup> cannot synthesise cross-modal evidence into unified reports. Clinical NLP systems such as ChatDoctor<sup>28</sup> and Med-PaLM<sup>29</sup> integrate RAG for general medical question answering but lack specialised cardiac signal processing modules.

### 5.3 | Limitations

1. **Inference latency:** The measured inference time of 57.39 s on Google Colab CPU substantially exceeds the real-time target of <2.0 s. GPU deployment on NVIDIA A100 hardware is expected to reduce latency below 2.0 s.
2. **Echocardiography demo mode:** The echo module was evaluated in synthetic demo mode due to EchoNet-Dynamic download constraints; prospective validation on real patient videos is required.
3. **GPT-2 limitations:** GPT-2 (124M parameters) is substantially smaller than current state-of-the-art language models; upgrading to a fine-tuned Llama-3-Med or BioMedLM may eliminate residual hallucinations observed in some generated reports.
4. **No prospective clinical validation:** The system is a research prototype not approved for clinical deployment.
5. **Demographic bias:** The NHANES arthritis dataset is a US-based national survey; generalisability to other ethnic populations and healthcare systems requires prospective multi-centre validation.

### 5.4 | Potential Applications

**Tele-cardiology and LMICs.** DeepCardio-RAG can provide preliminary screening reports from standard diagnostic inputs without specialist intervention, reducing diagnosis latency in resource-scarce settings.

**Second-opinion augmentation.** In high-resource settings, the platform can serve as a second-opinion tool, flagging discrepancies between AI and physician assessments.

**Continuous monitoring.** The modular architecture supports integration with wearable ECG devices and remote phonocardiography for continuous patient monitoring with automated alert generation.

**Medical education.** The dual-format PDF output has direct utility in medical education and patient health literacy programmes.

### 5.5 | Future Directions

1. **GPU optimisation:** Batched inference on A100 hardware targeting sub-2 s latency.
2. **Prospective clinical validation:** Multi-site trial with 500<sup>+</sup> real patient cases across diverse demographics.
3. **LLM upgrade:** Replace GPT-2 with Llama-3-Med or BioMedLM for superior, hallucination-free report generation.
4. **Linkage to richer clinical context:** none of the six corpora carry longitudinal notes, medications, or outcomes. Coupling the modules to an intensive-care database such as MIMIC-III<sup>59</sup> would allow the retrieval corpus to be grounded in patient history rather than guideline text alone.
5. **Federated learning:** Privacy-preserving multi-site training via federated averaging.
6. **Continual learning:** Online adaptation to emerging clinical guidelines without full retraining.
7. **Uncertainty quantification:** Bayesian deep learning or conformal prediction for calibrated confidence intervals.

## 6 | CONCLUSION

This paper has presented **DeepCardio-RAG v3.0**, a comprehensive multi-modal cardiac AI platform advancing the state of the art in three key dimensions.

**Multi-modal coverage:** By integrating six specialised deep-learning modules (1D-CNN Transformer for ECG, 3D-CNN for echocardiography, Mel-CNN for heart sounds, VFDB 1D-CNN for ventricular arrhythmia, and two TabularBERT-MoE models for heart disease and arthritis risk), the platform provides holistic cardiac assessment from a single unified inference pipeline—a capability not demonstrated by any prior system.

**RAG-grounded reporting:** The Milvus-backed RAG pipeline conditions GPT-2 generation on retrieved clinical guidelines via soft-prompt injection, providing evidence grounding for the generated report drafts. Quantitative evaluation of BLEU-4 and hallucination reduction against a reference corpus is identified as a key item for future work.

**Clinical accessibility:** Dual-format PDF generation—exercised end-to-end on PTB-XL record #10586—democratises access to specialist-quality cardiac assessments, with particular utility in tele-cardiology and resource-constrained settings.

Limitations, principally the current CPU-bound inference latency and absence of prospective clinical validation, are acknowledged with concrete pathways to address them. DeepCardio-RAG represents a meaningful step toward democratised, AI-assisted cardiology, with the potential to substantially improve diagnostic access and accuracy worldwide.

## ACKNOWLEDGEMENTS

The authors thank the open-source communities behind PyTorch, Milvus, and FastAPI, and the PhysioNet, EchoNet, CirCor, NHANES, and Kaggle dataset providers for making their data publicly available.



## APPENDIX

### A CORE PIPELINE PYTHON PSEUDOCODE

Listing 1: DeepCardio-RAG v3.0 – Core Pipeline

```

1 import torch, torch.nn as nn
2 from pymilvus import connections, Collection
3
4 class ECGEncoder(nn.Module):
5     # n_leads = 12 input channels (PTB-XL); C = 5 output classes
6     def __init__(self, n_leads=12, C=5, d=256):
7         super().__init__()
8         self.cnn = nn.Sequential(
9             nn.Conv1d(n_leads, 32, 5, padding=2), nn.BatchNorm1d(32),
10            nn.ReLU(), nn.MaxPool1d(2),
11            nn.Conv1d(32, 64, 5, padding=2), nn.BatchNorm1d(64),
12            nn.ReLU(), nn.MaxPool1d(2),
13            nn.Conv1d(64, 128, 5, padding=2), nn.BatchNorm1d(128),
14            nn.ReLU(), nn.MaxPool1d(2),
15            nn.Conv1d(128, 256, 5, padding=2), nn.BatchNorm1d(256),
16            nn.ReLU(), nn.MaxPool1d(2))
17        enc = nn.TransformerEncoderLayer(
18            d_model=d, nhead=8, dim_feedforward=1024)
19        self.transformer = nn.TransformerEncoder(enc, 6)
20        self.cls = nn.Parameter(torch.randn(1, 1, d))
21        self.head = nn.Linear(d, C)
22    def forward(self, x):
23        z = self.cnn(x).permute(2, 0, 1)
24        cls = self.cls.expand(-1, x.size(0), -1)
25        return self.head(self.transformer(
26            torch.cat([cls, z], 0)))[0])
27
28 class TabBERTMoE(nn.Module):
29     def __init__(self, F, K=6, d=64):
30         super().__init__()
31         self.embed = nn.ModuleList(
32             [nn.Linear(1, d) for _ in range(F)])
33         enc = nn.TransformerEncoderLayer(
34             d_model=d, nhead=4, dim_feedforward=256)
35         self.bert = nn.TransformerEncoder(enc, 4)
36         self.gate = nn.Linear(d, K)
37         self.experts = nn.ModuleList([
38             nn.Sequential(nn.Linear(d, 32), nn.ReLU(),
39                 nn.Linear(32, 1))
40             for _ in range(K)])
41    def forward(self, x):
42        E = torch.stack([self.embed[f](x[:, f:f+1])
43            for f in range(len(self.embed))], 1)
44        h0 = self.bert(E.permute(1, 0, 2))[0]
45        g = torch.softmax(self.gate(h0), -1)
46        out = sum(g[:, i:i+1]*self.experts[i](h0)
47            for i in range(len(self.experts)))
48        return torch.sigmoid(out)
49
50 def retrieve_guidelines(query_vec, collection, k=5):
51     connections.connect("default", host="localhost")
52     coll = Collection(collection)
53     res = coll.search(
54         data=[query_vec.tolist()],
55         anns_field="embedding",
56         param={"metric_type": "L2", "nprobe": 16},
57         limit=k, output_fields=["text"])
58     return [r.entity.get("text") for r in res[0]]
59

```



---

```

60 def cardiofusion(probs,
61                  weights=(0.25,0.25,0.20,0.15,0.10,0.05)):
62     score = 100*sum(w*p for w,p in zip(weights,probs))
63     level = ("LOW" if score<33 else
64             "MEDIUM" if score<66 else "HIGH")
65     return round(score,1), level

```

---

## References

1. World Health Organization: Cardiovascular diseases (CVDs). WHO Fact Sheet (2021). [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds))
2. Watkins, D.A., et al.: Cardiovascular and renal burdens of non-communicable diseases in low- and middle-income countries. *Lancet* **391**(10128), 1151–1161 (2018). [https://doi.org/10.1016/S0140-6736\(17\)32471-6](https://doi.org/10.1016/S0140-6736(17)32471-6)
3. Hannun, A.Y., et al.: Cardiologist-level arrhythmia detection and classification in ambulatory ECGs using a deep neural network. *Nat. Med.* **25**(1), 65–69 (2019). <https://doi.org/10.1038/s41591-018-0268-3>
4. Rajpurkar, P., et al.: Cardiologist-level arrhythmia detection with convolutional neural networks. arXiv:1707.01836 (2017)
5. Ouyang, D., et al.: Video-based AI for beat-to-beat assessment of cardiac function. *Nature* **580**(7802), 252–256 (2020). <https://doi.org/10.1038/s41586-020-2145-8>
6. Zhang, J., et al.: Fully automated echocardiogram interpretation in clinical practice. *Circulation* **138**(16), 1623–1635 (2018). <https://doi.org/10.1161/CIRCULATIONAHA.118.034338>
7. Madani, A., et al.: Fast and accurate view classification of echocardiograms using deep learning. *npj Digit. Med.* **1**, 6 (2018). <https://doi.org/10.1038/s41746-017-0013-1>
8. Leclerc, S., et al.: Deep learning for segmentation using an open large-scale dataset in 2D echocardiography. *IEEE Trans. Med. Imaging* **38**(9), 2198–2210 (2019). <https://doi.org/10.1109/TMI.2019.2900516>
9. Clifford, G.D., et al.: Classification of normal/abnormal heart sound recordings: The PhysioNet/CinC Challenge 2016. In: *Proceedings of Computing in Cardiology*, pp. 609–612 (2016)
10. Potes, C., et al.: Ensemble of feature-based and deep learning-based classifiers for detection of abnormal heart sounds. In: *Proceedings of Computing in Cardiology*, pp. 621–624 (2016)
11. Oliveira, J., et al.: The CirCor DigiScope dataset: From murmur detection to murmur classification. *IEEE J. Biomed. Health Inform.* **26**(6), 2524–2535 (2022). <https://doi.org/10.1109/JBHI.2021.3137048>
12. Acharya, U.R., Fujita, H., Oh, S.L., Hagiwara, Y., Tan, J.H., Adam, M., Tan, R.S.: Automated identification of shockable and non-shockable life-threatening ventricular arrhythmias using convolutional neural network. *Future Gener. Comput. Syst.* **79**, 952–959 (2018). <https://doi.org/10.1016/j.future.2017.08.039>
13. Acharya, U.R., et al.: A deep convolutional neural network for classification of heart sounds. *Comput. Methods Programs Biomed.* **152**, 1–10 (2017). <https://doi.org/10.1016/j.cmpb.2017.09.007>
14. Dang, H., et al.: A novel deep learning framework for automated detection of malignant ventricular arrhythmia. *IEEE Access* **7**, 59431–59440 (2019). <https://doi.org/10.1109/ACCESS.2019.2914352>
15. Pourbabaee, B., et al.: Deep convolutional neural networks and learning ECG features for screening paroxysmal atrial fibrillation. *IEEE Trans. Syst. Man Cybern.* **48**(12), 2095–2104 (2018). <https://doi.org/10.1109/TSMC.2017.2705582>
16. Ribeiro, A.H., et al.: Automatic diagnosis of the 12-lead ECG using a deep neural network. *Nat. Commun.* **11**, 1760 (2020). <https://doi.org/10.1038/s41467-020-15432-4>
17. Natarajan, A., et al.: A wide and deep transformer neural network for 12-lead ECG classification. In: *Proceedings of Computing in Cardiology* (2020). <https://doi.org/10.22489/CinC.2020.107>

18. Sun, C., Liu, C., Wang, X., Liu, Y., Zhao, S.: Coronary artery disease detection based on a novel multi-modal deep-coding method using ECG and PCG signals. *Sensors* **24**(21), 6939 (2024). <https://doi.org/10.3390/s24216939>
19. Shah, S.J., et al.: Cardiac risk assessment using machine learning. *J. Am. Coll. Cardiol.* **75**(25), 3167–3179 (2020). <https://doi.org/10.1016/j.jacc.2020.04.001>
20. Detrano, R., et al.: International application of a new probability algorithm for the diagnosis of coronary artery disease. *Am. J. Cardiol.* **64**(5), 304–310 (1989). [https://doi.org/10.1016/0002-9149\(89\)90524-9](https://doi.org/10.1016/0002-9149(89)90524-9)
21. Huang, X., et al.: TabTransformer: Tabular data modeling using contextual embeddings. *arXiv:2012.06678* (2020)
22. Shazeer, N., et al.: Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In: *Proceedings of ICLR* (2017)
23. Aviña-Zubieta, J.A., et al.: Risk of cardiovascular mortality in patients with rheumatoid arthritis: a meta-analysis of observational studies. *Arthritis Rheum.* **59**(12), 1690–1697 (2008). <https://doi.org/10.1002/art.24092>
24. Aletaha, D., et al.: 2010 rheumatoid arthritis classification criteria: ACR/EULAR collaborative initiative. *Arthritis Rheum.* **62**(9), 2569–2581 (2010). <https://doi.org/10.1002/art.27584>
25. Tian, J., et al.: Machine learning for prediction of rheumatoid arthritis risk from biomarkers. *J. Clin. Med.* **9**(4), 1233 (2020). <https://doi.org/10.3390/jcm9041233>
26. Lewis, P., et al.: Retrieval-augmented generation for knowledge-intensive NLP tasks. In: *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474 (2020)
27. Ji, Z., et al.: Survey of hallucination in natural language generation. *ACM Comput. Surv.* **55**(12), 1–38 (2023). <https://doi.org/10.1145/3571730>
28. Yunxiang, L., et al.: ChatDoctor: A medical chat model fine-tuned on LLaMA using medical domain knowledge. *arXiv:2303.14070* (2023)
29. Singhal, K., et al.: Large language models encode clinical knowledge. *Nature* **620**(7972), 172–180 (2023). <https://doi.org/10.1038/s41586-023-06291-2>
30. Nori, H., et al.: Capabilities of GPT-4 on medical challenge problems. *arXiv:2303.13375* (2023)
31. Wang, J., et al.: Milvus: A purpose-built vector data management system. In: *Proceedings of ACM SIGMOD*, pp. 2614–2627 (2021). <https://doi.org/10.1145/3448016.3457550>
32. Douze, M., et al.: The Faiss library. *arXiv:2401.08281* (2024)
33. Strodthoff, N., et al.: Deep learning for ECG analysis: Benchmarks and insights from PTB-XL. *IEEE J. Biomed. Health Inform.* **25**(5), 1519–1528 (2021). <https://doi.org/10.1109/JBHI.2020.3022989>
34. Wagner, P., Strodthoff, N., Bousseljot, R.-D., Kreiseler, D., Lunze, F.I., Samek, W., Schaeffter, T.: PTB-XL, a large publicly available electrocardiography dataset. *Sci. Data* **7**, 154 (2020). <https://doi.org/10.1038/s41597-020-0495-6>
35. Baloglu, U.B., et al.: Classification of myocardial infarction with multi-lead ECG signals and deep CNN. *Pattern Recognit. Lett.* **122**, 23–30 (2019). <https://doi.org/10.1016/j.patrec.2019.02.016>
36. Dezaki, F.T., et al.: Cardiac phase detection in echocardiograms using deep neural networks. *IEEE Trans. Ultrason. Ferroelectr. Freq. Control* **66**(3), 583–594 (2019). <https://doi.org/10.1109/TUFFC.2018.2889683>
37. Asch, F.M., et al.: Automated echocardiographic quantification of left ventricular ejection fraction. *Circ. Cardiovasc. Imaging* **12**(9), e009303 (2019). <https://doi.org/10.1161/CIRCIMAGING.119.009303>
38. Kay, M., et al.: Heart sound segmentation using recurrent neural networks. In: *Proceedings of EMBC*, pp. 2348–2352 (2020). <https://doi.org/10.1109/EMBC44109.2020.9176243>

39. Deng, X., et al.: Rethinking heart sound classification: Contrastive self-supervised learning. In: Proceedings of INTER-SPEECH, pp. 3571–3575 (2020)
40. Fallet, S., et al.: Multimodal signal quality index for automated ventricular fibrillation detection. In: Proceedings of Computing in Cardiology, pp. 857–860 (2015)
41. Shaik, K.A., et al.: Deep CNN for ventricular arrhythmia classification from ECG. *Biomed. Signal Process. Control* **68**, 102610 (2021). <https://doi.org/10.1016/j.bspc.2021.102610>
42. Weng, S.F., et al.: Can machine-learning improve cardiovascular risk prediction using routine clinical data? *PLOS ONE* **12**(4), e0174944 (2017). <https://doi.org/10.1371/journal.pone.0174944>
43. Krittanawong, C., et al.: Artificial intelligence in precision cardiovascular medicine. *J. Am. Coll. Cardiol.* **69**(21), 2657–2664 (2017). <https://doi.org/10.1016/j.jacc.2017.03.571>
44. England, B.R., et al.: Machine learning prediction of rheumatoid arthritis from routine blood tests. *Arthritis Res. Ther.* **20**, 228 (2018). <https://doi.org/10.1186/s13075-018-1705-9>
45. Ziegler, S., et al.: Cardiovascular risk attributable to rheumatoid arthritis. *Ann. Rheum. Dis.* **69**(11), 1949–1954 (2010). <https://doi.org/10.1136/ard.2009.122028>
46. Shi, W., et al.: REPLUG: Retrieval-augmented black-box language models. *arXiv:2301.12652* (2023)
47. Lester, B., et al.: The power of scale for parameter-efficient prompt tuning. In: Proceedings of EMNLP, pp. 3045–3059 (2021). <https://doi.org/10.18653/v1/2021.emnlp-main.243>
48. He, K., et al.: Deep residual learning for image recognition. In: Proceedings of CVPR, pp. 770–778 (2016). <https://doi.org/10.1109/CVPR.2016.90>
49. Vaswani, A., et al.: Attention is all you need. In: *Advances in Neural Information Processing Systems*, vol. 30, pp. 5998–6008 (2017)
50. Devlin, J., et al.: BERT: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of NAACL-HLT, pp. 4171–4186 (2019). <https://doi.org/10.18653/v1/N19-1423>
51. Radford, A., et al.: Language models are unsupervised multitask learners. *OpenAI Blog* **1**(8) (2019)
52. Lee, J., et al.: BioBERT: A pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics* **36**(4), 1234–1240 (2020). <https://doi.org/10.1093/bioinformatics/btz682>
53. Goodfellow, I., Bengio, Y., Courville, A.: *Deep Learning*. MIT Press, Cambridge (2016)
54. Litjens, G., et al.: A survey on deep learning in medical image analysis. *Med. Image Anal.* **42**, 60–88 (2017). <https://doi.org/10.1016/j.media.2017.07.005>
55. Shen, D., et al.: Deep learning in medical image analysis. *Annu. Rev. Biomed. Eng.* **19**, 221–248 (2017). <https://doi.org/10.1146/annurev-bioeng-071516-044442>
56. Topol, E.J.: High-performance medicine: The convergence of human and artificial intelligence. *Nat. Med.* **25**(1), 44–56 (2019). <https://doi.org/10.1038/s41591-018-0300-7>
57. Rajpurkar, P., et al.: AI in health and medicine. *Nat. Med.* **28**(1), 31–38 (2022). <https://doi.org/10.1038/s41591-021-01614-0>
58. Esteva, A., et al.: A guide to deep learning in healthcare. *Nat. Med.* **25**(1), 24–29 (2019). <https://doi.org/10.1038/s41591-018-0316-z>
59. Johnson, A.E.W., et al.: MIMIC-III, a freely accessible critical care database. *Sci. Data* **3**, 160035 (2016). <https://doi.org/10.1038/sdata.2016.35>
60. Goldberger, A.L., et al.: PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. *Circulation* **101**(23), e215–e220 (2000). <https://doi.org/10.1161/01.CIR.101.23.e215>

## STATEMENTS AND DECLARATIONS

**Funding.** The authors received no specific funding for this work. No funds, grants, or other support were received during the preparation of this manuscript.

**Competing Interests.** The authors have no relevant financial or non-financial interests to disclose. All authors certify that they have no affiliations with or involvement in any organisation or entity with any financial or non-financial interest in the subject matter or materials discussed in this manuscript.

**Author Contributions.** N.R.R. conceived and designed the study, developed the system architecture and ECG pipeline, supervised the project, and wrote the main manuscript text. D.N. implemented the software, integrated the multi-modal pipeline, conducted the experimental evaluation, and designed the clinical report generation module and dual-format PDF output. S.S. guided the arthritis and cardiac risk module design and contributed to the experimental methodology. N.S., as the clinical co-author, provided cardiology and clinical domain expertise, informed the clinical-safety framing and advisory-output design, validated the clinical plausibility of module outputs, and performed the medical literature review. All authors reviewed and approved the final manuscript.

**Data Availability.** All datasets used in this study are publicly available. MIT-BIH AAMI and MIT-BIH VFDB are available from PhysioNet (<https://physionet.org>); EchoNet-Dynamic from Stanford AIMI (<https://stanfordaimi.azurewebsites.net>); CirCor DigiScope 2022 from PhysioNet (<https://physionet.org/content/circor-heart-sound>); UCI Cleveland Heart Disease from the UCI Machine Learning Repository (<https://archive.ics.uci.edu>); and the NHANES Arthritis dataset from the CDC/Kaggle (<https://www.kaggle.com>). The prototype source code and supplementary materials are available from the corresponding author on reasonable request.